QUANT SERIES · PART 1 OF 5

Stochastic Calculus, Step by Step

Part 1: maths foundations,
from Taylor to the lognormal



Rates with Bip

STOCHASTIC CALCULUS, STEP BY STEP · PART 1 OF 5

Stochastic Calculus, Step by Step

Part 1: Maths foundations, with Python

Rates with Bip

Stochastic Calculus, Step by Step. Part 1: Maths foundations, with Python.

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One set of numbers runs through the whole course: a share priced at $S_0 = 100$, a strike $K = 105$, a risk-free rate $r = 3\%$ (continuously compounded), a volatility $\sigma = 20\%$ and a horizon $T = 1$ year. They are made up for teaching, and every random number is simulated with a fixed seed. There is no market data.

The exercises are at the end of each lesson. Their full solutions, and the answers to the self-test questions, are in the companion volume, Stochastic Calculus, Step by Step, Part 1: Solutions.

Every number in the text, tables and figures is computed by the companion Python notebooks (numpy only), one per lesson, which reproduce each result.

This book is for education only. It is not financial, investment, legal or tax advice.

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How to use this book

This is Part 1 of *Stochastic Calculus, Step by Step*, a course in five parts that goes from school calculus to the Black-Scholes formula, Feynman-Kac and the numerical methods banks use to price options. Part 1 lays the mathematical foundations: only what the later parts actually need, but each piece done properly.

LESSON	TOPIC	USED LATER FOR
1	Derivatives, the chain rule, Taylor's theorem, integrals, e and $\ln$	Itô's lemma, the Black-Scholes derivation
2	Partial derivatives, the two-variable Taylor expansion, $dm/dt = km$	Itô's lemma, $E[S^2]$
3	Random variables, densities, expectation, variance, covariance	everything
4	The normal distribution: moments, generating functions, completing the square	quadratic variation, the Black-Scholes formula, Girsanov
5	Lognormal prices, Jensen's inequality, the central limit theorem	geometric Brownian motion
6	The law of large numbers, Chebyshev's inequality, a convergence argument	quadratic variation, Monte Carlo

Who this book is for

This book is written for readers learning on their own, without a teacher to fill the gaps. So nothing is skipped. Every new idea is **defined** before it is used, **explained** in words (what each symbol means and why we want it), tried on a small example, and then **derived** one line at a time, with the reason for each line. When a step uses an earlier lesson, the lesson is named, so you can go back and check. All you need to start is school algebra: powers, functions and graphs.

How each lesson is built

Every lesson follows the same steps:

- **Why this lesson**: what the lesson is for and where it is used later.
- **What you need**: the facts from earlier lessons it relies on, collected in one place.

- **The picture**: a figure that shows the result before any algebra.
- **The derivations**, in full, as numbered steps, each with its reason and often a check.
- **Numbers and code**: every result tried on the course's numbers and checked in Python.
- **What to remember**: the lesson's results on one short list.
- **Common mistakes**, taken from questions that come up again and again.
- **Exercises**: eight per lesson, six by hand and two in code.
- **Self-test**: three quick questions to check you have the main ideas.

How to read it

Read with a pencil. When a derivation says "Step 2", try to do Step 2 yourself before reading it. Run the notebook next to the text and change the numbers. Then do the exercises before looking at the solutions; struggling with an exercise for a while is how the ideas stick.

One set of numbers

The whole course uses the same example: a share priced at $S_0 = 100$, a strike $K = 105$, a risk-free rate $r = 3\%$ (continuously compounded), a volatility $\sigma = 20\%$ and a horizon of $T = 1$ year. You'll meet the same numbers in every lesson, so each new tool is tried on familiar ground. They are made up for teaching; there is no market data in this book.

Notebooks and solutions

Every number, table and figure is computed by a companion Python notebook, one per lesson, using numpy only. The notebooks are free at github.com/RatesWithBip/curriculum.

The exercises are at the end of each lesson, and the solutions are **not** in this book: they are in the companion volume, *Part 1: Solutions*, with every derivation written out and every code solution shown with what it prints. Try each exercise before you look.

Reading alongside

Any first-year calculus text covers Lessons 1 and 2; for probability, Joseph Blitzstein and Jessica Hwang's *Introduction to Probability* (free online) and Thomas Mikosch's *Elementary Stochastic Calculus with Finance in View*, Chapter 1, are excellent. This book is written fresh from its own notebooks and uses none of their examples or exercises.

Free, and staying free

The videos, the notebooks and both books are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

This book is for education only and is not financial advice.

LESSON 1 · VIDEO 1

The Calculus You Need

What a derivative is and how to compute one; the rules, including the chain rule, and why they hold; the second derivative; Taylor's theorem, built up from a tangent line; why ordinary calculus drops the second-order term and why stochastic calculus can't; integrals, substitution and integration by parts; and the exponential and logarithm, with compounding and log returns.

Why this lesson

Stochastic calculus is ordinary calculus with one change: when the step is random, a term that ordinary calculus throws away has to be kept. That term is the second-order term of Taylor's theorem, and keeping it is all that **Itô's lemma** (Lesson 16) does.

So before anything random appears, this lesson rebuilds the calculus you need, from the definition of a derivative upwards:

- **derivatives** and the rules for computing them, especially the **chain rule**;
- **Taylor's theorem**, which approximates a function by a polynomial, and the question of when its second-order term can be dropped;
- **integrals**, with the two techniques used all through the course: **substitution** and **integration by parts**;
- the **exponential and logarithm**, which turn interest rates and share prices into formulas.

If you learned calculus at school, read quickly and do the exercises. If you never felt sure of it, read slowly: every rule is explained, not just stated.

What you need before starting

- **Functions and graphs.** A function f takes a number x and returns a number $f(x)$; its graph is the curve of points $(x, f(x))$.
- **Powers.** $x^0 = 1$, $x^{-n} = 1/x^n$, $x^{1/2} = \sqrt{x}$, $x^a x^b = x^{a+b}$, $(x^a)^b = x^{ab}$.

- **The idea of a limit.** $\lim_{h \to 0} g(h) = L$ means $g(h)$ gets as close as you like to L when h is close enough to 0 (but not equal to it). For example $\lim_{h \to 0}(2 + h) = 2$.

The picture

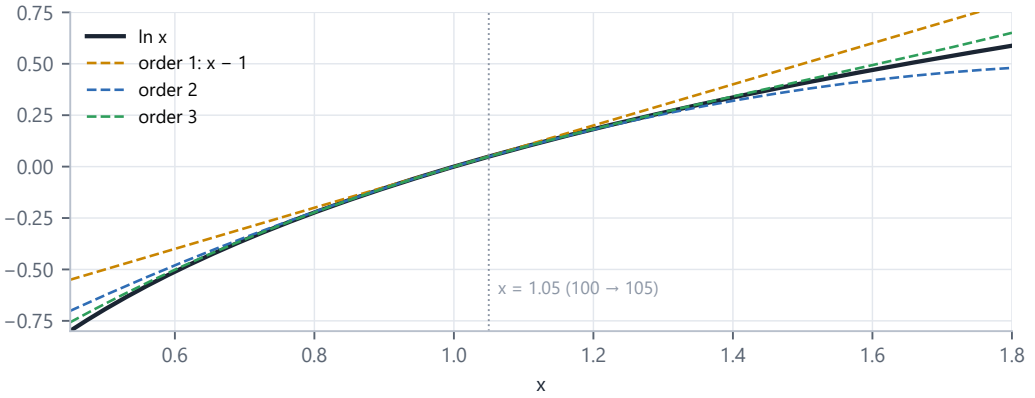


Figure 1.1. The logarithm and its Taylor polynomials around x = 1. Each extra order hugs the curve further out. Near x = 1, even the first-order line is close; at the dotted line, a 5% move, the second order is already very close.

Near the point where we start, every polynomial is a good copy of the curve; further away you need more terms. This lesson explains how the polynomials are built and when the second-order term can be dropped. The answer changes when the step is random, and that change is the whole of stochastic calculus.

The derivative

Slope of a curve

The slope of a straight line is "rise over run": how much it goes up per unit to the right. A curve has no single slope, but we can ask for its slope *at a point*.

Take two points on the graph of f: one at x and one a little to the right, at $x + h$. The straight line through them (a **secant**) has slope

$$\frac{f(x + h) - f(x)}{h} \qquad \text{(rise over run)}.$$

Now slide the second point towards the first, $h \to 0$. The secant turns into the **tangent**, the line that just touches the curve at x. Its slope is the derivative.

Definition. The **derivative** of f at x is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

when the limit exists. It is the rate at which f changes as x changes: if x moves by a small amount h, then f moves by about $f'(x)\,h$.

A first example from the definition

Take $f(x) = x^2$. Expand $(x+h)^2 = x^2 + 2xh + h^2$:

$$\frac{f(x+h) - f(x)}{h} = \frac{x^2 + 2xh + h^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h.$$

As $h \to 0$ this tends to $2x$, so $(x^2)' = 2x$. The same calculation with the binomial expansion gives the **power rule**, $(x^n)' = nx^{n-1}$, and it holds for any real power n, including $n = \frac{1}{2}$ (square roots) and negative n.

The rules

You will almost never use the definition directly. Instead, build every derivative out of a few rules:

RULE	FORMULA
Constant multiple	$(cf)' = cf'$
Sum	$(f+g)' = f' + g'$
Product	$(fg)' = f'g + fg'$
Chain rule	$\big(f(g(x))\big)' = f'(g(x))\,g'(x)$
Powers	$(x^n)' = nx^{n-1}$
Exponential	$(e^x)' = e^x$
Logarithm	$(\ln x)' = 1/x$

Why the product rule holds. Write $\Delta f = f(x+h) - f(x)$ and $\Delta g = g(x+h) - g(x)$. Then

$$f(x+h)g(x+h) - f(x)g(x) = \Delta f\, g(x) + f(x)\,\Delta g + \Delta f\,\Delta g.$$

(Check: expand $(f+\Delta f)(g+\Delta g) - fg$.) Divide by h and let $h \to 0$: the first two terms give $f'g + fg'$, and the last term is $\frac{\Delta f}{h}\Delta g \to f' \times 0 = 0$. $\square$

The chain rule

The chain rule is the rule you will use most. It handles a **function of a function**, such as e^{-rT} seen as a function of r: the inner function is $g(r) = -rT$ and the outer is $f(u) = e^u$.

Chain rule. If $y(x) = f(g(x))$, then $y'(x) = f'(g(x))\,g'(x)$: the derivative of the outer function, evaluated at the inner one, times the derivative of the inner one.

Why. Write $\Delta g = g(x+h) - g(x)$. Multiply and divide by Δg:

$$\frac{f(g(x+h)) - f(g(x))}{h} = \frac{f(g(x)+\Delta g) - f(g(x))}{\Delta g} \cdot \frac{\Delta g}{h}.$$

As $h \to 0$, $\Delta g \to 0$ too, so the first fraction tends to $f'(g(x))$ (it is the definition of f' with step Δg), and the second tends to $g'(x)$. (If Δg happens to be 0 for small h, both sides are 0 and the rule still holds.) □

In words: rates of change multiply. If u changes 3 times as fast as x, and f changes 2 times as fast as u, then f changes $2 \times 3 = 6$ times as fast as x.

Where e comes from, and why $(e^x)' = e^x$

For any base $b > 0$, the definition gives

$$\frac{b^{x+h} - b^x}{h} = b^x \cdot \frac{b^h - 1}{h}.$$

So the slope of b^x is b^x times the slope at 0, $\lim_{h\to 0}(b^h - 1)/h$. The number $e = 2.718282$ is *defined* as the base for which that slope at 0 is exactly 1. Then $(e^x)' = e^x \cdot 1 = e^x$: **the exponential is its own derivative**. You can see the slope at 0 approach 1 numerically:

(e^h − 1)/h approaches 1 as h shrinks.

STEP	SLOPE OF THE SECANT
0.1000	1.051709
0.0100	1.005017
0.0010	1.000500
0.0001	1.000050

The same number appears as a limit of compounding: $(1 + 1/n)^n$ is 2.5937 for $n = 10$, 2.7048 for $n = 100$ and 2.718146 for $n = 10{,}000$, approaching e. (Exercise 1.4 proves a version of this.)

Why $(\ln x)' = 1/x$. The logarithm undoes the exponential: $e^{\ln x} = x$ for $x > 0$. Differentiate both sides, using the chain rule on the left (outer e^u, inner $\ln x$):

$$e^{\ln x} \cdot (\ln x)' = 1 \quad\Longrightarrow\quad x \cdot (\ln x)' = 1 \quad\Longrightarrow\quad (\ln x)' = \frac{1}{x}.$$

Example: a bond's sensitivity to the interest rate

A zero-coupon bond pays 1 in T years. Discounted at the continuously compounded rate r, it is worth today

$$P(r) = e^{-rT}.$$

How fast does its price change when the rate changes? By the chain rule, with outer $f(u) = e^u$ ($f' = e^u$) and inner $g(r) = -rT$ ($g' = -T$):

$$P'(r) = e^{-rT} \cdot (-T) = -T\,e^{-rT}.$$

At $r = 3\%$, $T = 1$: $P = 0.970446$ and $P'(r) =$ -0.970446. The minus sign says the price falls when rates rise. A one-basis-point rise ($h = 0.0001$) lowers the price by about 0.0001×0.9704, which is about -9.7e-05.

Checking a derivative with a computer. If you are unsure of a derivative, compute the secant slope with a small h. The **forward difference** $\big(P(r+h) - P(r)\big)/h$ with $h = 10^{-4}$ is off by 4.9e-05; the **central difference** $\big(P(r+h) - P(r-h)\big)/(2h)$ is off by only -1.6e-09. Taylor's theorem, below, explains why the central one is so much better (Exercise 1.7).

The second derivative

The derivative f' is itself a function, so it has a derivative too: the **second derivative** $f'' = (f')'$. If f' is the slope, f'' is the rate at which the slope changes. It measures **curvature**:

- $f'' > 0$: the slope increases, the curve bends upwards (**convex**, like a bowl), as e^x and x^2 do;
- $f'' < 0$: the slope decreases, the curve bends downwards (**concave**, like a dome), as $\ln x$ and $\sqrt{x}$ do.

For the bond, differentiate $P'(r) = -T\,e^{-rT}$ once more with the chain rule:

$$P''(r) = -T \cdot e^{-rT} \cdot (-T) = T^2 e^{-rT} = 0.970446 > 0.$$

The bond price is a convex function of the rate. Traders call this **convexity**, and Exercise 1.6 shows what it does to a price change. Higher derivatives f''', f'''', $\dots$, $f^{(n)}$ are defined the same way.

Taylor's theorem

Building the polynomial

Suppose you know everything about f at one point a: its value, its slope, its curvature and so on. How well can you guess f at a nearby point $a + h$?

First guess: the tangent line. Keep the value and the slope:

$$f(a + h) \approx f(a) + f'(a)\, h.$$

This is the "moves by about $f'h$" rule from the definition of the derivative.

Better guess: add a parabola. Look for a polynomial in h,

$$p(h) = c_0 + c_1 h + c_2 h^2 + c_3 h^3 + \dots,$$

that matches f and its derivatives at $h = 0$. Differentiate p repeatedly and set $h = 0$:

- $p(0) = c_0$, so matching the value needs $c_0 = f(a)$;
- $p'(h) = c_1 + 2c_2 h + 3c_3 h^2 + \dots$, so $p'(0) = c_1$ and we need $c_1 = f'(a)$;
- $p''(h) = 2c_2 + 3 \cdot 2\, c_3 h + \dots$, so $p''(0) = 2c_2$ and we need $c_2 = f''(a)/2$;
- $p'''(0) = 3 \cdot 2 \cdot 1\, c_3$, so $c_3 = f'''(a)/3!$;
- in general $c_k = f^{(k)}(a)/k!$, where $k! = 1 \times 2 \times \cdots \times k$.

That polynomial is the **Taylor polynomial**. Taylor's theorem says how close it is.

Theorem (Taylor, with remainder). If f has $n + 1$ continuous derivatives near a, then

$$f(a + h) = f(a) + f'(a)\, h + \frac{f''(a)}{2!} h^2 + \cdots + \frac{f^{(n)}(a)}{n!} h^n + R_n(h),$$

where the remainder is

$$R_n(h) = \frac{f^{(n+1)}(\xi)}{(n+1)!} h^{n+1} \quad \text{for some } \xi \text{ between } a \text{ and } a + h.$$

The remainder looks exactly like the next term, except that the derivative is taken at an unknown point ξ instead of at a.

Why, for $n = 1$. By the fundamental theorem of calculus (below), $f(a + h) - f(a) = \int_a^{a+h} f'(s)\, ds$. Integrate by parts (also below) with $u = f'(s)$ and $dv = ds$, choosing $v = -(a + h - s)$:

$$\int_a^{a+h} f'(s)\, ds = \Big[-(a + h - s) f'(s) \Big]_a^{a+h} + \int_a^{a+h} (a + h - s)\, f''(s)\, ds.$$

The bracket is $0 - (-h\,f'(a)) = f'(a)\,h$. In the last integral, $(a + h - s) \geq 0$ (take $h > 0$), so the integral lies between $\min f''$ and $\max f''$ times $\int_a^{a+h} (a + h - s)\,ds = h^2/2$. A continuous function takes every value between its minimum and maximum, so the integral equals $f''(\xi)\,h^2/2$ for some ξ. Integrating by parts again gives each further term the same way. □

What the remainder tells you. If $f^{(n+1)}$ stays bounded near a, the error after the h^n term is at most a constant times $|h|^{n+1}$. So halving h cuts the first-order error about $2^2 = 4$ times and the second-order error about $2^3 = 8$ times.

Example: the exponential series

Every derivative of e^x is e^x, which is 1 at $a = 0$. So all the coefficients are $1/k!$:

$$e^h = 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \cdots + \frac{h^n}{n!} + R_n(h).$$

The remainder is $e^{\xi} h^{n+1}/(n+1)!$, and $(n+1)!$ grows so fast that it goes to 0 as $n \to \infty$ for every h. So the series converges, $e^h = \sum_{k=0}^{\infty} h^k/k!$. Lesson 4 uses this series to build the moment generating function.

Example: a 5% move and the logarithm

A move from 100 to 105 is a relative change $h = 0.05$, and its **log return** is $\ln(105/100) = \ln(1 + h)$. Use $f(x) = \ln x$ at $a = 1$. The derivatives are

$$f'(x) = \frac{1}{x}, \qquad f''(x) = -\frac{1}{x^2}, \qquad f'''(x) = \frac{2}{x^3},$$

so at $x = 1$: $f = 0$, $f' = 1$, $f''/2! = -\frac{1}{2}$, $f'''/3! = \frac{2}{6} = \frac{1}{3}$. Hence

$$\ln(1 + h) = h - \frac{h^2}{2} + \frac{h^3}{3} - \ldots$$

ORDER	APPROXIMATION	ERROR
exact	0.048790	
1	0.050000	1.2e-03
2	0.048750	-4.0e-05
3	0.048792	1.5e-06

Each extra term cuts the error by a factor of about 20 here. Halving the move confirms the orders: as h goes from 0.1 to 0.05 to 0.025, the first-order error falls by factors of 3.88 and 3.94 (close to 4), and the second-order error by 7.72 and 7.86 (close to 8).

The three Taylor approximations of ln(1 + h), from the notebook.

```
h = K / S0 - 1
exact = math.log(1 + h)
orders = {"1": h, "2": h - h**2 / 2, "3": h - h**2 / 2 + h**3 / 3}
out["taylor"] = {"h": h, "exact": exact, **{f"o{k}": v for k, v in orders.items()},
```

The first-order result is worth remembering: **for small moves, the log return and the simple return are almost equal**, $\ln(1+h) \approx h$, and the log return is a little smaller, by about $h^2/2$.

Why the second-order term usually goes, and when it can't

This section is the reason the lesson exists.

Many small steps in ordinary calculus

Ordinary calculus builds a change over a long time out of many small steps. Split one year into n equal steps of length $h = 1/n$, and suppose that over each step x changes by an amount of size h (smooth motion). By Taylor, each step changes f by

$$\underbrace{f' h}_{\text{first order}} + \underbrace{\tfrac{1}{2} f'' h^2}_{\text{second order}} + \dots$$

Now add up the n steps:

- The first-order terms add up to about $n \times f'h = f' \times nh = f' \times 1$: something finite.
- The second-order terms add up to about $n \times \frac{1}{2} f'' h^2 = \frac{1}{2} f'' \times nh \times h = \frac{1}{2} f'' \, h$, which goes to 0 as $h \to 0$.

So in the limit, **the second-order terms disappear**, and only first derivatives matter. That is why school calculus never needs f'' to describe how things change.

Random steps in finance

A share price does not move smoothly. Over a short time Δt, a typical relative move is not of size Δt but of size $\sigma\sqrt{\Delta t}$ (Lesson 6 shows why; Lesson 14 makes it precise). Since $\sqrt{\Delta t}$ is much bigger than Δt when Δt is small, the moves are rough. Over one trading day, $\Delta t = 1/252$:

ONE TRADING DAY	SIZE
typical relative move $\sigma\sqrt{\Delta t}$	0.01260
its half-square $\frac{1}{2}(\sigma\sqrt{\Delta t})^2$	7.9e-05
the drift $r\,\Delta t$	1.2e-04

The second-order term is now **the same size as the drift**: both are proportional to Δt. Add up the year again. The squares of the 252 daily moves add up to

$$252 \times \left(\sigma\sqrt{\Delta t}\right)^2 = 252 \times \sigma^2 \Delta t = \sigma^2 = 0.04,$$

which does **not** vanish however finely you split the year. For comparison, the squares of smooth steps of size $r\Delta t$ add up to only 3.6e-06.

So when the steps are random, the second-order Taylor term must be kept. Itô's lemma (Lesson 16) is Taylor's theorem with exactly this term kept. Everything else in stochastic calculus follows from it.

Integrals

The integral as an area

For a function $f \geq 0$ on $[a, b]$, cut the interval into thin slices of width Δx at points x_i. Each slice is roughly a rectangle of height $f(x_i)$, with area $f(x_i)\,\Delta x$. The **integral** is the limit of the total as the slices get thinner:

$$\int_a^b f(x)\,dx = \lim_{\Delta x \to 0} \sum_i f(x_i)\,\Delta x.$$

(Where $f < 0$ the area counts as negative.) The notation remembers the sum: $\int$ is a stretched S for "sum", and dx is the slice width.

The fundamental theorem: integrals undo derivatives

Computing that limit directly is hard. The **fundamental theorem of calculus** turns it into a derivative problem: if $F' = f$, then

$$\int_a^b f(x)\,dx = F(b) - F(a), \qquad \text{written } \Big[F(x)\Big]_a^b.$$

Why. Let $A(x) = \int_a^x f(s)\,ds$, the area from a to x. Increasing x by h adds a thin slice of width h and height about $f(x)$, so $A(x+h) - A(x) \approx f(x)\,h$, and $A'(x) = f(x)$. Two functions with the same derivative differ by a constant, so $A(x) = F(x) + c$; since $A(a) = 0$, $c = -F(a)$, and $A(b) = F(b) - F(a)$. $\square$

An F with $F' = f$ is called an **antiderivative**. Reading the derivative rules backwards gives $\int x^n\,dx = x^{n+1}/(n+1)$ (for $n \neq -1$), $\int e^x\,dx = e^x$ and $\int \frac{1}{x}\,dx = \ln x$.

Example. One unit of money a year, paid continuously over a year and growing at rate r, accumulates to

$$\int_0^1 e^{rt}\,dt = \left[\frac{e^{rt}}{r}\right]_0^1 = \frac{e^r - 1}{r} = 1.015151.$$

(Check the antiderivative with the chain rule: $\frac{d}{dt}\frac{e^{rt}}{r} = \frac{re^{rt}}{r} = e^{rt}$.) Adding up 10,000 thin slices with the trapezoid rule gives 1.015151.

Substitution: the chain rule backwards

The chain rule says $\frac{d}{dx}F(g(x)) = F'(g(x))\,g'(x)$. Integrating both sides gives the **substitution rule**:

$$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

The recipe.

1. Spot an inner function $u = g(x)$ whose derivative $g'(x)$ also appears in the integrand.
2. Write $du = g'(x)\,dx$.
3. **Change the limits**: $x = a$ becomes $u = g(a)$, and $x = b$ becomes $u = g(b)$.
4. Integrate in u.

Example. Compute $\int_0^\infty x\,e^{-x^2/2}\,dx$.

1. The inner function is $u = -x^2/2$, and its derivative $-x$ almost appears: we have $x\,dx$.
2. $du = -x\,dx$, so $x\,dx = -du$.
3. When $x = 0$, $u = 0$; when $x \to \infty$, $u \to -\infty$.
4. So

$$\int_0^\infty x\,e^{-x^2/2}\,dx = \int_0^{-\infty} e^u\,(-du) = \int_{-\infty}^0 e^u\,du = \Big[e^u\Big]_{-\infty}^0 = 1 - 0 = 1.$$

(Swapping the limits removed the minus sign.) The trapezoid rule gives 1.000000. Lesson 4 uses this integral to find the average size of a normal random variable.

Integration by parts: the product rule backwards

The product rule says $(uv)' = u'v + uv'$. Integrate both sides from a to b and rearrange:

$$\int_a^b u\,v'\,dx = \Big[u\,v\Big]_a^b - \int_a^b u'\,v\,dx.$$

When to use it. When the integrand is a product, and differentiating one factor (u) makes it simpler while the other (v') is easy to integrate.

Example. Compute $\int_0^1 x\,e^x\,dx$. Take $u = x$ (so $u' = 1$, simpler) and $v' = e^x$ (so $v = e^x$):

$$\int_0^1 x\,e^x\,dx = \Big[x\,e^x\Big]_0^1 - \int_0^1 e^x\,dx = (e - 0) - (e - 1) = 1.$$

The trapezoid rule gives 1.000000. Lesson 4 uses integration by parts to show that the variance of a standard normal is 1, and the proof of Taylor's theorem above used it too.

Exponentials and logarithms

The rules, and why they hold

e^x and $\ln x$ undo each other: $e^{\ln x} = x$ (for $x > 0$) and $\ln(e^x) = x$. The rules:

$$e^{a+b} = e^a e^b, \qquad \ln(ab) = \ln a + \ln b, \qquad \ln\frac{a}{b} = \ln a - \ln b, \qquad \ln(a^b) = b\ln a.$$

The logarithm rules follow from the exponential ones. For example, $e^{\ln a + \ln b} = e^{\ln a}e^{\ln b} = ab$; taking logarithms of both sides gives $\ln a + \ln b = \ln(ab)$. **The logarithm turns products into sums**, and that is why it is everywhere in finance, where returns multiply.

Continuous compounding

Invest 100 at an annual rate r compounded n times a year: each period multiplies by $(1 + r/n)$, so after a year you have $100(1 + r/n)^n$. More frequent compounding pays interest on interest sooner:

Compounding 3% more and more often.

TIMES A YEAR	100 AFTER ONE YEAR
1	103.0000
2	103.0225
4	103.0339
12	103.0416
52	103.0446
365	103.0453

As $n \to \infty$ the amount tends to $100e^{r}$ (Exercise 1.4 proves it): this is **continuous compounding**, and it is the convention in this course. Here $100e^{0.03} = 103.0455$, against 103.00 for simple interest.

Running the clock backwards gives **discounting**: an amount X paid in T years is worth Xe^{-rT} today, because Xe^{-rT} invested now grows to $Xe^{-rT}e^{rT} = X$. Discounting the strike: $105e^{-0.03} = 101.8968$.

Log returns add up; simple returns don't

The **simple return** from S_0 to S_1 is $S_1/S_0 - 1$, and the **log return** is $\ln(S_1/S_0)$.

Suppose the share goes 100, 110, 99. The simple returns are +10% and −10%. They add to 0, but the share ended at 99, down 1%: simple returns do not add up.

The log returns are $\ln(110/100) = 0.0953$ and $\ln(99/110) =$ -0.1054. They add to -0.0101, which is exactly the whole period's log return $\ln(99/100)$. The reason is the logarithm rule:

$$\ln\frac{S_1}{S_0} + \ln\frac{S_2}{S_1} = \ln\left(\frac{S_1}{S_0}\cdot\frac{S_2}{S_1}\right) = \ln\frac{S_2}{S_0}.$$

Log returns over consecutive periods add up to the log return over the whole period. That is why price models are written in log returns, and why Lesson 4's result that sums of independent normals are normal applies directly to them.

What to remember

- $f'(x) = \lim_{h\to 0}\big(f(x+h) - f(x)\big)/h$: the slope, and the rate of change. f'' is the curvature.
- Chain rule: $\big(f(g(x))\big)' = f'(g(x))\,g'(x)$. $(e^x)' = e^x$, $(\ln x)' = 1/x$.

- Taylor: $f(a+h) = f(a) + f'(a)h + \frac{1}{2}f''(a)h^2 + \ldots$; the error after the h^n term is of size h^{n+1}.
- $e^h = \sum h^k/k!$ and $\ln(1+h) = h - h^2/2 + h^3/3 - \ldots$
- Over many smooth steps the second-order terms vanish; over random steps of size $\sigma\sqrt{\Delta t}$ they add up to $\sigma^2 T$ and must be kept.
- $\int_a^b f = F(b) - F(a)$ with $F' = f$. Substitution (change the limits!) and integration by parts.
- Continuous compounding: $100 \to 100e^{rT}$; discounting: $X \to Xe^{-rT}$. Log returns add.

Common mistakes

- **"$\ln(a+b) = \ln a + \ln b$."** No: the logarithm turns *products* into sums. $\ln(100 + 5) \neq \ln 100 + \ln 5$.
- **Forgetting the inner derivative in the chain rule.** $(e^{-rT})' = e^{-rT}$ is wrong when differentiating in r: the inner function $-rT$ contributes the factor $-T$.
- **Forgetting to change the limits in a substitution.** After $u = g(x)$ the limits must be $g(a)$ and $g(b)$, not a and b.
- **"The second-order term can always be dropped."** Only when the step is small *and* its square is much smaller than the step. For a random step of size $\sigma\sqrt{\Delta t}$ the square is of size Δt, like the drift, and must be kept.
- **"e^{rT} and $1 + rT$ are the same thing."** Close for small rT, but not equal: $100e^{0.03} = 103.0455$ against 103, and the gap grows with rT.

Exercises

Solutions are in *Part 1: Solutions*.

1.1 Differentiate $f(x) = e^{-x^2/2}$ and $g(S) = \ln(Se^{rT})$. What is $g'(100)$?

1.2 Write the second-order Taylor polynomial of $\sqrt{x}$ around $x = 1$, and use it to approximate $\sqrt{1.21}$. How big is the error?

1.3 Compute $\int_0^T e^{-rt}\,dt$, the value today of 1 a year paid continuously for T years. Evaluate it for $r = 3\%$, $T = 1$.

1.4 Show that $(1 + r/n)^n \to e^r$ as $n \to \infty$, using $\ln(1+x) = x - \frac{1}{2}x^2 + \ldots$.

1.5 Use integration by parts to compute $\int_0^T t\,e^{-rt}\,dt$. Evaluate it for $r = 3\%$, $T = 1$.

1.6 For the bond price $P(r) = e^{-rT}$ with $T = 1$, find $P''(r)$. Estimate the price change when the rate rises from 3% to 4% with the first-order and the second-order Taylor approximations, and compare both with the exact change.

1.7 *(Code)* For $f(x) = e^x$ at $x = 0$, compute the forward-difference and central-difference errors for $h = 10^{-1}, 10^{-2}, \ldots, 10^{-10}$. Why does each error first fall and then rise again?

1.8 *(Code)* A share goes 100, 110, 99, 108.9. Compute the simple and log returns of each step, and show which kind adds up to the whole period's return.

Self-test

1. Why can ordinary calculus drop the second-order Taylor term, and what changes when the step is a random move of size $\sigma\sqrt{\Delta t}$?
2. What is $100e^{0.03}$ to two decimals, and how far is it from 100×1.03?
3. If $\ln(S_T/S_0) = 0.05$, what was the simple return?

Companion notebook: `s5e01_calculus_refresher.ipynb`. Every number in this chapter is computed by it.

LESSON 2 · VIDEO 2

Multivariable Calculus

Functions of two variables and what their graphs look like; partial derivatives, how to compute them and what they mean for an option; the multivariable chain rule; the two-variable Taylor expansion with its cross term, and which of its terms survive when one step is random; and the equation dm/dt = km, solved exactly and by Euler's method.

Why this lesson

An option's value depends on two things at once: the **time** t and the **share price** S. As the day goes by, both change. How much does the option's value change?

Lesson 1 answered the one-variable version of that question with Taylor's theorem. This lesson does the same for two variables. You will need:

- **partial derivatives**, which measure the change in one variable at a time (these are the option "Greeks" theta, delta and gamma);
- the **multivariable chain rule**, which adds the effects when both variables move along a path;
- the **two-variable Taylor expansion**, including the **cross term** that is easy to forget, and the question of which second-order terms can be dropped when the price step is random. **Itô's lemma** (Lesson 16) is exactly this expansion with the right terms kept.

The lesson ends with the one differential equation that keeps coming back in later parts: $dm/dt = km$.

What you need from Lesson 1

- Derivatives, the product rule and the chain rule.
- Taylor's theorem in one variable: $f(a+h) = f(a) + f'(a)h + \frac{1}{2}f''(a)h^2 + \dots$, with an error of size h^3 after the h^2 term.
- Over a short time Δt, a random price move is of size $\sqrt{\Delta t}$, so its square is of size Δt.

Functions of two variables

A **function of two variables** $g(t, x)$ takes two numbers and returns one. Examples:

- the value of an option, $V(t, S)$, depending on the time and the share price;
- this lesson's running example, $g(t, x) = e^{-rt}x^2$: a share price x, squared and discounted back from time t.

Its graph is a **surface**: above each point (t, x) of the flat plane, at height $g(t, x)$.

The picture

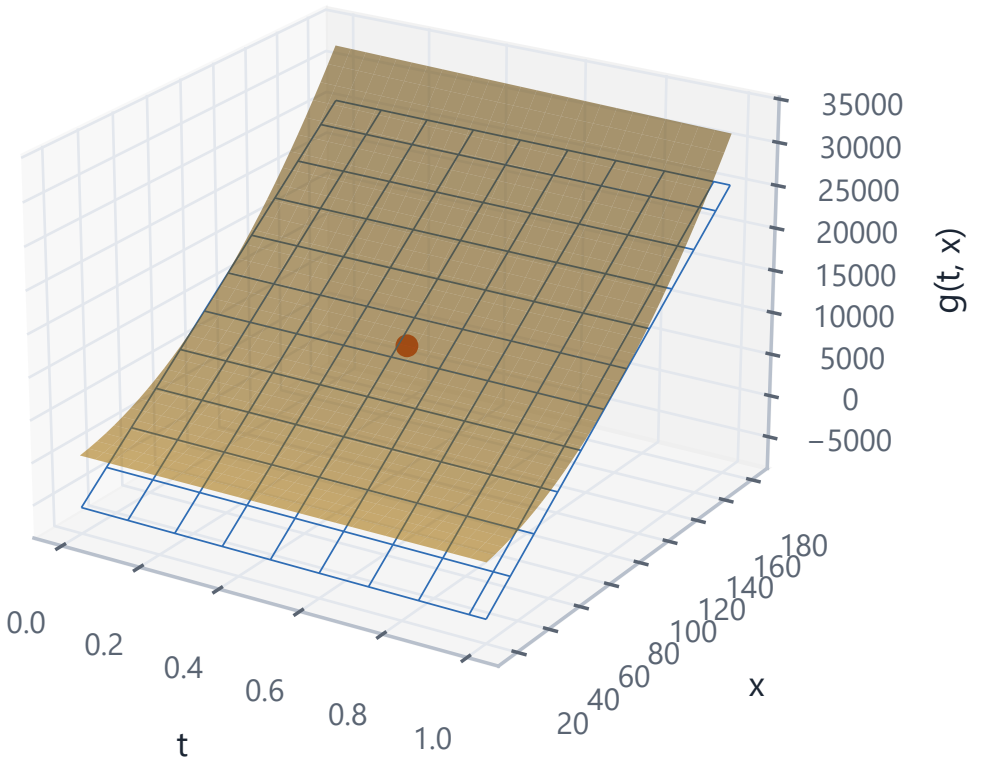


Figure 2.1. The surface g(t, x) = e^(−rt) x^2 and its tangent plane (blue mesh) at t = 0.5, x = 100 (red dot). Near the dot the plane is a good fit; away from it the surface curves upwards in the x direction.

Stand on the surface at the red dot. Walking in the x direction you climb steeply and the ground curves upwards; walking in the t direction the ground falls gently. The two **partial derivatives** are the slopes in those two directions, and the flat **tangent plane** built

from them fits the surface near the dot. The second derivatives say how fast the surface curves away from the plane.

Partial derivatives

Definition

Definition. The **partial derivative of g with respect to x** is the ordinary derivative in x with t held fixed:

$$g_x(t,x) = \frac{\partial g}{\partial x}(t,x) = \lim_{h \to 0} \frac{g(t,x+h) - g(t,x)}{h}.$$

Likewise g_t is the derivative in t with x held fixed.

The curly ∂ (read "partial") is a reminder that only one variable is moving. Geometrically, g_x is the slope of the curve you get by slicing the surface at a fixed time t.

How to compute it. Treat the other variable as a constant, like the number 7, and use Lesson 1's rules.

Second derivatives

Each partial derivative is again a function of (t,x), so it has partial derivatives of its own:

- $g_{xx} = (g_x)_x$: how the slope in x changes as x changes, the curvature in the x direction;
- $g_{tt} = (g_t)_t$: the curvature in the t direction;
- the **mixed** derivatives $g_{tx} = (g_t)_x$ and $g_{xt} = (g_x)_t$.

Theorem (symmetry of mixed derivatives). If g_{tx} and g_{xt} are continuous, they are equal. So the order does not matter, and we write g_{tx} for both. (Every function in this course is smooth enough. The proof uses the mean value theorem twice; we take it as a fact and check it on examples.)

Worked example

Take $g(t,x) = e^{-rt}x^2$.

g_t: hold x fixed, so x^2 is a constant multiplier. The derivative of e^{-rt} in t is $-re^{-rt}$ (chain rule):

$$g_t = -re^{-rt}x^2.$$

g_x: hold t fixed, so e^{-rt} is a constant multiplier. The derivative of x^2 is $2x$:

$$g_x = 2e^{-rt}x.$$

Second derivatives. Differentiate again, each time holding the other variable fixed:

$$g_{xx} = 2e^{-rt}, \qquad g_{tt} = r^2e^{-rt}x^2.$$

Mixed derivatives, both ways. Differentiate $g_t = -re^{-rt}x^2$ in x: $g_{tx} = -2re^{-rt}x$. Differentiate $g_x = 2e^{-rt}x$ in t: $g_{xt} = -2re^{-rt}x$. They agree, as the theorem says.

Numbers. At $(t, x) = (0.5, 100)$:

DERIVATIVE	FORMULA	VALUE
g	$e^{-rt}x^2$	9851.12
g_t	$-re^{-rt}x^2$	-295.53
g_x	$2e^{-rt}x$	197.0224
g_{xx}	$2e^{-rt}$	1.970224
g_{tx}	$-2re^{-rt}x$	-5.910672
g_{tt}	$r^2e^{-rt}x^2$	8.8660

Central finite differences, as in Lesson 1, agree with all of them to better than 1 part in 10^5 (the notebook checks it).

What they mean for an option

If $V(t, S)$ is an option's value, its partial derivatives have names that traders use every day:

- V_S is **delta**: the change in value per unit change in the share price;
- V_{SS} is **gamma**: how fast delta itself changes;
- V_t is **theta**: the change in value per unit of time passing, with the share price unchanged.

Lesson 21 computes them for the Black-Scholes formula. The Black-Scholes equation (Lesson 19) is a relation between exactly these three.

The tangent plane

In one variable, $f(a + h) \approx f(a) + f'(a)h$. In two variables, move t by Δt and x by Δx and think of it as two steps: first in t, then in x. Each step changes g by about its slope times its size, so

$$\Delta g = g(t + \Delta t, x + \Delta x) - g(t, x) \approx g_t\,\Delta t + g_x\,\Delta x.$$

The right-hand side, as a function of $(\Delta t, \Delta x)$, is the tangent plane of the picture. It is the first-order Taylor approximation; we will add the second-order terms below.

The multivariable chain rule

Now suppose x is not free but follows a path in time, $x = x(t)$, for example the share price over the day. Then $g(t, x(t))$ is a function of t alone. What is its derivative?

Theorem. If $x(t)$ is differentiable and g has continuous partial derivatives, then

$$\frac{d}{dt}\, g\big(t, x(t)\big) = g_t\big(t, x(t)\big) + g_x\big(t, x(t)\big)\, x'(t).$$

In words: the total rate of change is the direct effect of time, g_t, plus the effect through the moving price: the slope g_x times the speed $x'(t)$ at which the price moves.

Why. Over a step Δt, x changes by $\Delta x = x(t+\Delta t) - x(t)$. Split the change of g into a step in t (at the new x) and a step in x:

$$\Delta g = \underbrace{\big[g(t+\Delta t, x+\Delta x) - g(t, x+\Delta x)\big]}_{\text{only } t \text{ moves}} + \underbrace{\big[g(t, x+\Delta x) - g(t, x)\big]}_{\text{only } x \text{ moves}}.$$

Each bracket is a one-variable change. The **mean value theorem** (Lesson 1's Taylor theorem with $n = 0$: $f(a+h) - f(a) = f'(\xi)h$ for some ξ between a and $a+h$) turns the first bracket into $g_t \times \Delta t$ and the second into $g_x \times \Delta x$, each evaluated at some nearby point. Divide by Δt:

$$\frac{\Delta g}{\Delta t} = g_t(\text{nearby}) + g_x(\text{nearby})\, \frac{\Delta x}{\Delta t}.$$

As $\Delta t \to 0$ the nearby points tend to (t, x), and $\Delta x / \Delta t \to x'(t)$. $\square$

Example. Let the price grow like money in the bank, $x(t) = S_0 e^{rt}$, so $x'(t) = rS_0 e^{rt} = r\,x(t)$. With $g = e^{-rt}x^2$:

Step 1. Chain rule.

$$\frac{d}{dt} g = g_t + g_x\, x' = -re^{-rt}x^2 + 2e^{-rt}x \cdot rx = (-r + 2r)\, e^{-rt}x^2 = re^{-rt}x^2.$$

Step 2. Check by substituting first. $g(t, x(t)) = e^{-rt}S_0^2 e^{2rt} = S_0^2 e^{rt}$, whose derivative is $rS_0^2 e^{rt}$. With $x = S_0 e^{rt}$, $re^{-rt}x^2 = rS_0^2 e^{rt}$: the same.

Step 3. Numbers. At $t = 0.5$ both give 304.5339, and a finite difference gives 304.5339.

Substituting first is fine when the path is a known formula. The chain rule is what you need when it isn't, and Itô's lemma is the chain rule for a random path.

Two-variable Taylor, with the cross term

The expansion

Theorem. If g has continuous third partial derivatives near (t, x), then

$$\Delta g = g_t\,\Delta t + g_x\,\Delta x + \tfrac{1}{2}\,Q + R,$$

where the second-order part is

$$Q = g_{tt}\,\Delta t^2 + 2g_{tx}\,\Delta t\,\Delta x + g_{xx}\,\Delta x^2,$$

all derivatives are taken at (t, x), and the remainder $|R|$ is at most a constant times $(|\Delta t| + |\Delta x|)^3$.

Why: reduce to one variable. Walk along the straight line from (t, x) to $(t + \Delta t, x + \Delta x)$. Let

$$\phi(s) = g(t + s\,\Delta t,\ x + s\,\Delta x), \qquad 0 \le s \le 1,$$

so $\phi(0) = g(t, x)$ and $\phi(1) = g(t + \Delta t, x + \Delta x)$. This is a function of one variable, s, so Lesson 1's Taylor theorem applies with $a = 0$ and $h = 1$:

$$\phi(1) - \phi(0) = \phi'(0) + \tfrac{1}{2}\phi''(0) + (\text{third-order remainder}).$$

We only need ϕ' and ϕ''.

Step 1. First derivative. Along the line, t moves at speed Δt and x at speed Δx per unit of s. By the chain rule (with both variables moving),

$$\phi'(s) = g_t\,\Delta t + g_x\,\Delta x.$$

Step 2. Second derivative. Apply the same rule to each of the two terms. Differentiating $g_t\,\Delta t$ gives $(g_{tt}\,\Delta t + g_{tx}\,\Delta x)\,\Delta t$, and differentiating $g_x\,\Delta x$ gives $(g_{xt}\,\Delta t + g_{xx}\,\Delta x)\,\Delta x$. Add them:

$$\phi''(s) = g_{tt}\,\Delta t^2 + g_{tx}\,\Delta t\,\Delta x + g_{xt}\,\Delta x\,\Delta t + g_{xx}\,\Delta x^2 = Q,$$

using $g_{tx} = g_{xt}$.

Step 3. Put $\phi'(0)$ and $\phi''(0)$ into the one-variable Taylor formula. □

The cross term $2g_{tx}\,\Delta t\,\Delta x$ comes from both variables moving together. Its factor 2 is there because the mixed derivative appears **twice** in Step 2, once from each first-order term. Forgetting the 2, or the whole term, is a classic mistake.

Which terms matter over one day?

Take $g(t, x) = e^{-rt}x^2$ at $(0.5, 100)$, one trading day $\Delta t = 1/252$, and a typical one-day price move $\Delta x = \sigma x\sqrt{\Delta t} = 1.2599$. Here are the five terms:

The five Taylor terms for one trading day and a typical one-day move.

TERM	SIZE
g_t dt	-1.172752
g_x dx	248.224877
1/2 g_tt dt^2	0.000070
g_tx dt dx	-0.029551
1/2 g_xx dx^2	1.563670

The actual change is 248.5861. The first-order approximation misses it by -1.5340; adding the second-order terms brings the miss down to 1.8e-04.

Look at the sizes. The term $\frac{1}{2}g_{xx}\Delta x^2$ is 1.33 times as big as the time term $g_t\Delta t$, while $\frac{1}{2}g_{tt}\Delta t^2$ and the cross term are tiny. The reason is how big each kind of step is when Δx is of size $\sqrt{\Delta t}$:

With Δx of size √Δt, Δx^2 is as big as Δt, while Δt^2 and Δt Δx are much smaller.

QUANTITY	SIZE AS A POWER OF ΔT	ONE DAY
Δt	1	0.003968
Δx ≈ √Δt	1/2	0.062994
Δt²	2	0.000016
Δt Δx	3/2	0.000250
Δx²	1	0.003968

As $\Delta t \to 0$, the terms of size Δt survive when added up over many steps (as in Lesson 1), and the terms of size $\Delta t^{3/2}$ or Δt^2 vanish. So for a random price the rule is:

$$\Delta g \approx g_t\,\Delta t + g_x\,\Delta x + \tfrac{1}{2}g_{xx}\,\Delta x^2.$$

Keep the first-order terms and the Δx^2 term; drop Δt^2 and $\Delta t\,\Delta x$. Lesson 16 shows that, added up over many steps, Δx^2 can even be replaced by $\sigma^2 x^2\,\Delta t$, and that is Itô's lemma.

The five terms, from the notebook.

```
dt = 1 / 252
dx = sigma * x0 * math.sqrt(dt)
actual = g(t0 + dt, x0 + dx) - g(t0, x0)
terms = {"g_t dt": exact["gt"] * dt, "g_x dx": exact["gx"] * dx, "1/2 g_tt dt^2": 0.5 *
exact["gtt"] * dt**2,
         "g_tx dt dx": exact["gtx"] * dt * dx, "1/2 g_xx dx^2": 0.5 * exact["gxx"] * dx**2}
first = terms["g_t dt"] + terms["g_x dx"]
```

The equation $dm/dt = km$

What a differential equation is

A **differential equation** is an equation for an unknown *function*, involving its derivatives. The simplest one used in finance says that a quantity grows at a rate proportional to its current size:

$$m'(t) = k\,m(t), \qquad m(0) = m_0.$$

Money in a bank account at continuous rate k does this: interest per unit of time is k times the balance. The second condition, the **initial condition**, fixes where it starts.

The solution

Proposition. The solution is $m(t) = m_0 e^{kt}$, and it is the only one.

It works. By the chain rule, $\frac{d}{dt} m_0 e^{kt} = k m_0 e^{kt}$, which is k times the function; and at $t = 0$ it equals m_0.

It is the only one. Let m be any solution. Multiply it by e^{-kt} and differentiate with the product rule:

$$\frac{d}{dt}\big(m(t)\,e^{-kt}\big) = m'(t)\,e^{-kt} - k\,m(t)\,e^{-kt} = \big(m'(t) - k\,m(t)\big)e^{-kt} = 0.$$

A function with derivative 0 is constant, so $m(t)e^{-kt}$ always equals its value at $t = 0$, which is m_0. Hence $m(t) = m_0 e^{kt}$. $\square$

Multiplying by e^{-kt} is the **integrating factor** trick. It also solves $m' = km + c$ (Exercise 2.4), and Part 3 uses it again for interest-rate models.

Euler's method: solving it step by step

When a differential equation has no formula for its solution, a computer can step it forward. **Euler's method** uses the first-order Taylor approximation over a small step Δt:

$$m(t + \Delta t) \approx m(t) + m'(t)\,\Delta t = m(t) + k\,m(t)\,\Delta t.$$

Start at m_0, apply this n times with $\Delta t = T/n$, and you get an approximation of $m(T)$.

Where it comes back. In Lesson 17, $m(t) = E[S_t^2]$ for a share following geometric Brownian motion turns out to satisfy $m' = (2r + \sigma^2)\, m$. With $k = 2r + \sigma^2 = 0.10$ and $m_0 = S_0^2 = 10{,}000$, the exact answer is $m(1) = m_0 e^k = 11051.7092$. Euler's method converges to it:

Euler's method for m′ = 0.1 m, m(0) = 10,000, over one year.

STEPS	EULER	ERROR
1	11000.0000	-51.7092
10	11046.2213	-5.4879
100	11051.1570	-0.5522
1,000	11051.6539	-0.0553

Each tenfold increase in steps cuts the error about tenfold: Euler's method has error proportional to $1/n$ (Exercise 2.8 explains why). The error is always negative here, because the tangent line lies below the convex curve e^{kt}.

What to remember

- g_x: differentiate in x with t held fixed. Mixed derivatives don't depend on the order.
- Delta, gamma, theta are V_S, V_{SS}, V_t.
- Chain rule along a path: $\frac{d}{dt} g(t, x(t)) = g_t + g_x\, x'(t)$.
- Taylor: $\Delta g = g_t \Delta t + g_x \Delta x + \frac{1}{2}\left(g_{tt}\Delta t^2 + 2g_{tx}\Delta t \Delta x + g_{xx}\Delta x^2\right) + \dots$
- When Δx is of size $\sqrt{\Delta t}$: keep $g_t \Delta t$, $g_x \Delta x$ and $\frac{1}{2} g_{xx} \Delta x^2$; drop the rest.
- $m' = km, m(0) = m_0 \Rightarrow m(t) = m_0 e^{kt}$. Euler: $m \leftarrow m + km\,\Delta t$, error proportional to Δt.

Common mistakes

- **Forgetting the cross term,** or forgetting its factor 2. In ϕ'' the mixed derivative appears twice.
- **Not holding the other variable fixed.** In $\partial/\partial x\, (e^{-rt}x^2)$, the factor e^{-rt} is a constant; it does not get differentiated.
- **"A partial derivative is the total change."** g_x freezes t. Along a path both variables move, and the chain rule adds both effects.
- **Dropping the wrong second-order term.** Which second-order terms are small depends on how big the steps are relative to each other. With Δx of size $\sqrt{\Delta t}$, Δx^2

stays and Δt^2 goes.

Exercises

Solutions are in *Part 1: Solutions*.

2.1 For $f(t,x) = \ln x + (r - \tfrac{1}{2}\sigma^2)(T-t)$, compute f_t, f_x and f_{xx}.

2.2 Let $f(t,x) = t\,x^2$ and $x(t) = e^{at}$. Find $\frac{d}{dt} f(t, x(t))$ twice: with the chain rule, and by substituting first.

2.3 Expand $f(x,y) = xy$ around $(1,2)$ to second order. Show that the expansion is exact, and that dropping the cross term leaves an error of exactly $\Delta x\,\Delta y$.

2.4 Solve $m'(t) = k\,m(t) + c$ with $m(0) = m_0$. Evaluate $m(1)$ for $k = 0.1$, $c = 5$, $m_0 = 100$.

2.5 Find all the first and second partial derivatives of $f(t,x) = t\,x^2 + e^{tx}$, check that $f_{tx} = f_{xt}$, and evaluate f_{tx} at $(t,x) = (1, 0.5)$.

2.6 Show that $f(t,x) = e^{x - t/2}$ and $g(t,x) = x^2 - t$ both satisfy $f_t + \tfrac{1}{2} f_{xx} = 0$. For which constant c does $x^3 - c\,t\,x$ satisfy it too?

2.7 *(Code)* For $g(t,x) = e^{-rt}x^2$ at $(0.5, 100)$, halve $(\Delta t, \Delta x)$ repeatedly and show that the second-order Taylor error falls about 8 times each time.

2.8 *(Code)* Solve $m' = km$ with $k = 0.10$, $m(0) = 10{,}000$ by Euler steps, $n = 10, 100, 1000, 10{,}000$. Show that the error falls in proportion to $1/n$.

Self-test

1. In the one-day example, which second-order term is largest, and why is it so much bigger than the other two?
2. Solve $m' = -0.03\,m$ with $m(0) = 105$. What is $m(1)$, and what is it in finance terms?
3. A long forward contract struck at K is worth $f(t,x) = x - Ke^{-r(T-t)}$. What are f_x and f_t, and what do they say?

Companion notebook: `s5e02_multivariable_calculus.ipynb`. Every number in this chapter is computed by it.

LESSON 3 · VIDEO 3

Probability Basics

Outcomes, events and the rules of probability; random variables and their distributions; expectation, first for a list of outcomes and then with a density; the formula E[g(X)] = ∫ g f and why E[g(X)] is not g(E[X]); linearity, variance and standard deviation; covariance, correlation and independence. Every price in this course is an expectation.

Why this lesson

Every price in this course will be written as an **expectation**: the average of a payoff over the possible futures, weighted by how likely each future is. Part 2 will make the weights clever (the "risk-neutral" probabilities). Before that, the averages themselves must be second nature.

This lesson builds them from the start:

- what a **random variable** is and what its **distribution** is;
- how to compute an **expectation**, for a list of outcomes and for a continuous price with a **density**;
- the rules that make expectations easy to work with: **linearity**, the **variance** formula;
- how two random quantities can be related: **covariance**, **correlation** and **independence**.

What you need from Lesson 1

- Integrals as areas, $\int_a^b f(x)\,dx = F(b) - F(a)$ with $F' = f$.
- Substitution, and the power rule for integrals, $\int x^n\,dx = x^{n+1}/(n+1)$.

The picture

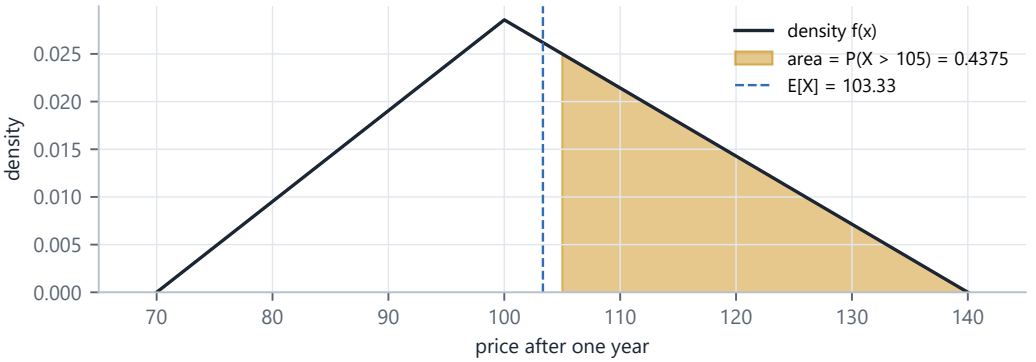


Figure 3.1. A triangular density for next year's price, from 70 to 140, peaking at 100. Probabilities are areas: the shaded area is the chance the price ends above the strike, 105. The dashed line is the mean.

This curve describes an uncertain share price a year from now. The price will be somewhere between 70 and 140, most likely near 100. The rule for reading it is simple: **probabilities are areas under the curve**. The total area is 1 (the price will be *somewhere*), and the shaded area is the chance it ends above 105. By the end of the lesson you will compute that area, the average price (the dashed line) and the average payoff of a call option.

Outcomes, events and probabilities

Think of the future as one of many possible **outcomes**: all the ways the world might turn out. The set of all of them is written Ω (omega).

An **event** is a statement about the outcome that is either true or false, such as "the share ends above 105". Formally it is the set of outcomes for which it is true.

A **probability** P gives each event a number between 0 and 1, following three rules:

1. $0 \le P(A) \le 1$ for every event A;
2. $P(\Omega) = 1$: something happens;
3. if A and B cannot both happen, $P(A \text{ or } B) = P(A) + P(B)$.

Two consequences used constantly:

- **The complement rule.** "Not A" and A cannot both happen and together cover everything, so $P(A) + P(\text{not } A) = 1$, that is, $P(\text{not } A) = 1 - P(A)$.
- **Splitting.** If outcomes are a list of separate cases, the probability of an event is the sum of the probabilities of the cases where it is true.

(Part 2 makes "which events have a probability" precise, with σ-algebras. For now, every event you can describe in words has one.)

Random variables and their distributions

Definition. A **random variable** is a number that depends on the outcome. We write it with a capital letter, X or S; the values it can take are written with small letters.

The share price in a year, S, is a random variable: different futures give different prices. So is a call's payoff, $\max(S - 105, 0)$: once you know the price, you know the payoff. **A function of a random variable is another random variable.**

The **distribution** of X says how likely each value is. When X takes a list of values $x_1, x_2, \ldots$, with probabilities $p_i = P(X = x_i)$ adding to 1, X is **discrete** and its distribution is just that table.

Example: a share with three outcomes. In a year the share is worth 80, 100 or 125:

PRICE	PROBABILITY
80	0.25
100	0.45
125	0.30

The probabilities add to 1, as they must.

Expectation of a discrete random variable

Definition. The **expectation** (or **mean**, or **expected value**) of a discrete random variable is the probability-weighted average of its values:

$$E[X] = \sum_i x_i \, p_i .$$

Two ways to think about it:

- **Long-run average.** If you could run the year many times, the average of the prices you saw would settle near $E[X]$ (Lesson 6 proves this: the law of large numbers).
- **Balance point.** Put weights p_i at positions x_i on a ruler; $E[X]$ is where the ruler balances.

Expectation of a function. To average $g(X)$, average the values $g(x_i)$ with the same probabilities:

$$E[g(X)] = \sum_i g(x_i)\, p_i.$$

Worked example

The mean. Multiply each price by its probability and add:

$$E[S] = 80(0.25) + 100(0.45) + 125(0.30) = 20 + 45 + 37.5 = 102.50.$$

A probability. The share ends above the strike 105 only in the outcome 125, so $P(S > 105) = 0.30$.

A call's expected payoff. A call with strike 105 pays $g(S) = \max(S - 105, 0)$. Work out the payoff in each outcome first:

PRICE	CALL PAYOFF	PROBABILITY
80	0	0.25
100	0	0.45
125	20	0.30

Then average: $E[\max(S - 105, 0)] = 0(0.25) + 0(0.45) + 20(0.30) = 6.00$.

That is the shape of every option price in this course: list the outcomes, compute the payoff in each, and average with the right probabilities (plus discounting, from Part 2 on).

Continuous random variables and densities

Why a density is needed

A share price can take any value in a range, not just three. With infinitely many possible values, each single value has probability 0: the chance the price is exactly 103.000000... is zero. So we cannot list $P(X = x)$. Instead we describe how probability is **spread out** along the line.

Definition. A random variable X has a **density** f if $f(x) \geq 0$, $\int_{-\infty}^{\infty} f(x)\, dx = 1$, and for all $a < b$

$$P(a < X < b) = \int_a^b f(x)\, dx.$$

How to read a density. Over a thin slice from x to $x + dx$, the area is about $f(x)\, dx$, so

$$P(x < X < x + dx) \approx f(x)\, dx.$$

The height $f(x)$ is not itself a probability: it is **probability per unit length**, like population per square kilometre. It can be bigger than 1, if the slice is narrow enough. Only areas are probabilities.

The distribution function. $F(x) = P(X \le x) = \int_{-\infty}^{x} f(s)\,ds$, the area to the left of x. By the fundamental theorem of calculus, $F' = f$. And $P(a < X < b) = F(b) - F(a)$.

Expectation with a density

Replace the sum by an integral. Each thin slice near x has probability about $f(x)\,dx$, so the probability-weighted average of the values is

$$E[X] = \int_{-\infty}^{\infty} x\, f(x)\,dx, \qquad E[g(X)] = \int_{-\infty}^{\infty} g(x)\, f(x)\,dx.$$

The second formula is important enough to have a name, the **law of the unconscious statistician**. It says you never need the distribution of $g(X)$ itself: just weight $g(x)$ by the density of X. That is how every option price is computed: g is the payoff, f the density of the price.

The triangular price, worked in full

Let next year's price have the triangular density of the picture: zero outside $[70, 140]$, rising in a straight line from 70 to a peak at 100, and falling in a straight line to 140.

Step 1. The height of the peak. The area of a triangle is $\frac{1}{2} \times$ base $\times$ height. The base is $140 - 70 = 70$, and the area must be 1, so $\frac{1}{2} \times 70 \times h = 1$ and $h = 1/35 = 0.028571$.

Step 2. The formula for f. Each side is a straight line.

- Left side, from $(70, 0)$ up to $(100, \frac{1}{35})$: slope $\frac{1/35}{30} = \frac{1}{1050}$, so $f(x) = \dfrac{x - 70}{1050}$ for $70 \le x \le 100$.
- Right side, from $(100, \frac{1}{35})$ down to $(140, 0)$: slope $-\frac{1/35}{40} = -\frac{1}{1400}$, so $f(x) = \dfrac{140 - x}{1400}$ for $100 \le x \le 140$.

(Check at the peak: $30/1050 = 40/1400 = 1/35$.)

Step 3. A probability. The chance the price ends above the strike is the shaded area. For $x \ge 105$ we are on the right side:

$$P(X > 105) = \int_{105}^{140} \frac{140 - x}{1400}\,dx.$$

Substitute $u = 140 - x$, $du = -dx$; $x = 105$ gives $u = 35$ and $x = 140$ gives $u = 0$:

$$\int_{105}^{140} \frac{140 - x}{1400}\, dx = \frac{1}{1400}\int_{0}^{35} u\, du = \frac{1}{1400}\cdot\frac{35^2}{2} = \frac{1225}{2800} = 0.4375.$$

(It is also the area of the small triangle: base 35, height $35/1400$.)

Step 4. The mean. Split the integral at the peak:

$$E[X] = \int_{70}^{100} x\,\frac{x - 70}{1050}\, dx + \int_{100}^{140} x\,\frac{140 - x}{1400}\, dx.$$

In the first integral substitute $u = x - 70$ (so $x = u + 70$, u from 0 to 30):

$$\frac{1}{1050}\int_{0}^{30} (u + 70)\, u\, du = \frac{1}{1050}\left[\frac{u^3}{3} + 35u^2\right]_0^{30} = \frac{9000 + 31{,}500}{1050} = 38.5714.$$

In the second substitute $u = 140 - x$ (so $x = 140 - u$, u from 40 down to 0, and the minus sign of $du = -dx$ flips the limits back):

$$\frac{1}{1400}\int_{0}^{40} (140 - u)\, u\, du = \frac{1}{1400}\left[70u^2 - \frac{u^3}{3}\right]_0^{40} = \frac{112{,}000 - 21{,}333.33}{1400} = 64.7619.$$

Adding: $E[X] = 38.5714 + 64.7619 = 103.3333$. (For any triangle the answer is the average of the three corners, $(70 + 100 + 140)/3$; the calculation above shows why it works here.) The mean is above the peak at 100 because the right side stretches further.

Step 5. The call's expected payoff. With $g(x) = \max(x - 105, 0)$, which is 0 below 105:

$$E[\max(X - 105, 0)] = \int_{105}^{140} (x - 105)\,\frac{140 - x}{1400}\, dx.$$

Substitute $u = x - 105$ (so $140 - x = 35 - u$, u from 0 to 35):

$$\frac{1}{1400}\int_{0}^{35} u\,(35 - u)\, du = \frac{1}{1400}\left[\frac{35u^2}{2} - \frac{u^3}{3}\right]_0^{35} = \frac{35^3}{1400}\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{35^3}{8400} = 5.1042.$$

The notebook checks each integral numerically, and with a million simulated prices (the simulated call average is 5.0985).

$E[g(X)]$ is not $g(E[X])$

A tempting shortcut is to compute the payoff at the average price. The average price is 103.33, below the strike, so the call on the average price is worth $\max(103.33 - 105, 0) = 0.00$. But the **average payoff** is 5.1042.

The shortcut throws away exactly what makes the option valuable: the outcomes above 105, where it pays. For a straight-line g the shortcut works (that is linearity, next); for a bent payoff it doesn't. Lesson 5 turns this into **Jensen's inequality**.

Linearity of expectation

Rule. For any random variables X, Y and numbers a, b, c:

$$E[aX + bY + c] = aE[X] + bE[Y] + c.$$

Why, for one variable. For a discrete X, using $\sum_i p_i = 1$:

$$E[aX + c] = \sum_i (ax_i + c)\, p_i = a \sum_i x_i p_i + c \sum_i p_i = aE[X] + c.$$

With a density, the same steps work with integrals, because integrals are linear too. The two-variable version needs the joint distribution of X and Y, but the idea is the same: an average of a sum is the sum of the averages. It holds **whether or not X and Y are related**, and that is what makes it so useful.

Variance and standard deviation

The mean says where a random variable is centred. We also need to know how spread out it is.

Definition. The **variance** is the average squared distance from the mean:

$$\mathrm{Var}(X) = E\big[(X - \mu)^2\big], \qquad \mu = E[X].$$

The **standard deviation** is its square root, $\mathrm{sd}(X) = \sqrt{\mathrm{Var}(X)}$.

Why squares? The plain distances $X - \mu$ average to 0 (by linearity, $E[X - \mu] = \mu - \mu = 0$): ups and downs cancel. Squaring makes every distance count positively, and squares behave well in algebra, as the next rules show. The square root at the end brings the units back: the variance of a price in dollars is in dollars squared, the standard deviation is in dollars.

A shortcut formula. Expand the square and use linearity, remembering that μ is a number:

$$\begin{aligned} \mathrm{Var}(X) &= E\big[X^2 - 2\mu X + \mu^2\big] \\ &= E[X^2] - 2\mu\, E[X] + \mu^2 \\ &= E[X^2] - 2\mu^2 + \mu^2 = E[X^2] - \mu^2. \end{aligned}$$

So $\mathrm{Var}(X) = E[X^2] - E[X]^2$: "the mean of the square minus the square of the mean".

Example. For the three-outcome share, $E[S^2] = 6400(0.25) + 10{,}000(0.45) + 15{,}625(0.30) = 10787.50$ and $E[S]^2 = 102.5^2 = 10{,}506.25$, so

$$\mathrm{Var}(S) = 10787.50 - 10{,}506.25 = 281.25, \qquad \mathrm{sd}(S) = 16.7705.$$

For the triangular price, the same method (longer arithmetic) gives $\mathrm{Var}(X) = 205.5556$ and $\mathrm{sd}(X) = 14.3372$.

Scaling and shifting. For numbers a and b, $E[aX + b] = a\mu + b$, so

$$\mathrm{Var}(aX + b) = E\big[(aX + b - a\mu - b)^2\big] = E\big[a^2(X - \mu)^2\big] = a^2\,\mathrm{Var}(X).$$

Shifting by b does not change the spread; scaling by a multiplies the standard deviation by $|a|$. **Example:** two shares of the triangular price plus 50 in cash, $2X + 50$, have mean $2 \times 103.33 + 50 = 256.67$ and variance $4 \times 205.56 = 822.22$.

Covariance and correlation

When two random quantities are in play, such as two share prices, we need to know whether they tend to move together.

Definition. The **covariance** of X and Y is

$$\mathrm{Cov}(X, Y) = E\big[(X - E[X])(Y - E[Y])\big].$$

How to read it. When X and Y are both above their means, or both below, the product inside is positive; when one is above and the other below, it is negative. So $\mathrm{Cov} > 0$ means they tend to move together, $\mathrm{Cov} < 0$ that they tend to move in opposite directions.

Shortcut. Expanding the product and using linearity, exactly as for the variance:

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]\,E[Y].$$

Note that $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$.

The variance of a sum. Write $\tilde{X} = X - E[X]$ and $\tilde{Y} = Y - E[Y]$, so that $X + Y - E[X + Y] = \tilde{X} + \tilde{Y}$. Then

$$\mathrm{Var}(X + Y) = E\big[(\tilde{X} + \tilde{Y})^2\big] = E[\tilde{X}^2] + E[\tilde{Y}^2] + 2E[\tilde{X}\tilde{Y}],$$

that is,

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y).$$

Correlation. Covariance depends on units (dollars times dollars). Dividing by both standard deviations gives a pure number, the **correlation**:

$$\rho(X,Y) = \frac{\operatorname{Cov}(X,Y)}{\operatorname{sd}(X)\operatorname{sd}(Y)}.$$

It always lies between -1 and 1. It is 1 when Y is an increasing straight-line function of X, -1 for a decreasing one, and 0 when there is no linear relation. **Example:** the notebook builds a second price that moves with the triangular one, $Y = 0.5(X - E[X]) +$ noise; their covariance is 103.07 and their correlation 0.58.

Independence

Definition. X and Y are **independent** if knowing one tells you nothing about the other:

$$P(X \in A \text{ and } Y \in B) = P(X \in A)\,P(Y \in B) \quad \text{for all sets } A, B.$$

Independence makes products easy. If X and Y are independent, then $E[XY] = E[X]\,E[Y]$.

Why, for discrete variables. With $P(X = x_i, Y = y_j) = p_i q_j$:

$$E[XY] = \sum_i \sum_j x_i y_j\, p_i q_j = \Big(\sum_i x_i p_i\Big)\Big(\sum_j y_j q_j\Big) = E[X]\,E[Y].$$

Functions of independent variables are again independent, so also $E[g(X)h(Y)] = E[g(X)]\,E[h(Y)]$. Lesson 4 uses this with g and h exponentials.

Consequences. Independent X and Y have $\operatorname{Cov}(X,Y) = E[XY] - E[X]E[Y] = 0$, so **their variances add**:

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) \qquad \text{(independent)}.$$

This is why, in later lessons, the variance of a sum of independent daily returns grows in proportion to the number of days.

The converse fails. Zero covariance does *not* imply independence. Let U be uniform on $[-1, 1]$ and $V = U^2$. Then $E[U] = 0$ and $E[UV] = E[U^3] = 0$ (both by symmetry), so $\operatorname{Cov}(U,V) = 0 - 0 \cdot E[V] = 0$ (simulated: -3.2e-05). Yet V is completely determined by U: whenever $|U| > 0.5$, $V > 0.25$ for certain, while overall $P(V > 0.25) = 0.50$. Covariance only measures **linear** dependence, and U^2 is not a straight-line function of U.

Zero covariance, complete dependence.

```
U = rng.uniform(-1, 1, 1_000_000)
V = U**2
Y = 0.5 * (X - tri["mean_exact"]) + rng.normal(0, 10, X.size)        # a second price that
moves with X
```

What to remember

- A random variable is a number that depends on the outcome; its distribution says how likely each value is.
- Discrete: $E[g(X)] = \sum g(x_i)p_i$. With a density: $P(a < X < b) = \int_a^b f$, $E[g(X)] = \int g(x)f(x)\,dx$.
- A density is probability per unit length; only areas are probabilities. $F(x) = P(X \le x)$, $F' = f$.
- $E[g(X)] \neq g(E[X])$ unless g is a straight line.
- Linearity: $E[aX + bY + c] = aE[X] + bE[Y] + c$, always.
- $\mathrm{Var}(X) = E[X^2] - E[X]^2$, $\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$.
- $\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$; $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$; $\rho = \mathrm{Cov}/(\mathrm{sd}\,\mathrm{sd})$.
- Independent $\Rightarrow E[XY] = E[X]E[Y]$ and variances add. Zero covariance does not imply independence.

Common mistakes

- "$E[g(X)] = g(E[X])$." Only for straight-line g. For a call payoff the difference is the whole option value.
- **"A density is a probability."** $f(x)$ is probability per unit length; it can exceed 1. Probabilities are areas, $\int f$.
- **Forgetting that the variance shortcut squares the mean.** $\mathrm{Var}(X) = E[X^2] - E[X]^2$, not $E[X^2] - E[X]$.
- **"Zero covariance means independent."** It means no *linear* relation; U and U^2 are the standard counterexample.
- **"Variances always add."** Only when the covariance is 0. Two shares that move together have $\mathrm{Var}(X + Y) > \mathrm{Var}(X) + \mathrm{Var}(Y)$.
- **"Standard deviations add."** Even for independent variables, it is the *variances* that add.

Exercises

Solutions are in *Part 1: Solutions*.

3.1 For the three-outcome share, compute $E[S]$, $\mathrm{Var}(S)$, $P(S > 105)$ and $E[\max(S - 105, 0)]$.

3.2 If X is uniform on $[80, 130]$, show that $E[\max(X - K, 0)] = (130 - K)^2/100$ for $80 \le K \le 130$. Evaluate it at $K = 105$.

3.3 Show that $\mathrm{Var}(2X + 50) = 4\,\mathrm{Var}(X)$ and $\mathrm{Cov}(X, 2X + 50) = 2\,\mathrm{Var}(X)$.

3.4 Let Z take the values -1, 0, 1 with probability $\frac{1}{3}$ each, and $Y = Z^2$. Show that $\mathrm{Cov}(Z, Y) = 0$ but Z and Y are not independent.

3.5 For the triangular density, find the distribution function $F(x) = P(X \le x)$ on both sides of the peak, check that $F(100) = 3/7$ and $F(140) = 1$, and use F to compute $P(90 < X < 110)$.

3.6 Two share prices X and Y each have standard deviation 20, and their correlation is ρ. Find the standard deviation of $X + Y$ for $\rho = 1$, 0.5, 0 and -1. What does this say about holding both?

3.7 *(Code)* Draw 1,000,000 samples from the triangular density (by inverse transform) and check $E[X]$, $\mathrm{Var}(X)$ and $E[\max(X - 105, 0)]$ against the exact values.

3.8 *(Code)* Build two correlated prices and check $\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X, Y)$ by simulation.

Self-test

1. For the triangular price, what are $E[\max(X - 105, 0)]$ and $\max(E[X] - 105, 0)$? Why do they differ?
2. If $\mathrm{Cov}(X, Y) = 0$, are X and Y independent?
3. X and Y are independent with standard deviations 3 and 4. What is the standard deviation of $X - Y$?

Companion notebook: `s5e03_probability_basics.ipynb`. Every number in this chapter is computed by it.

LESSON 4 · VIDEO 4

The Normal Distribution

What the normal distribution is and where its $\sqrt{(2\pi)}$ comes from; standardising and symmetry; moments and the moment generating function (what it is, why we use it, and how to compute it); the average of an exponential of a normal; sums of independent normals; completing the square, all the way to the first term of the Black-Scholes formula; and E|Z|.

Why this lesson

Almost everything later in this course is built out of normal random variables:

- **Brownian motion** (Lessons 13 to 18) is a path whose steps are independent normals.
- **The Black-Scholes formula** (Lesson 20) is written with the normal distribution function N.
- **Girsanov's theorem** (Lesson 25) is a statement about what happens when you multiply a normal density by an exponential.

So this lesson takes the normal distribution apart slowly. We start from its formula, check every property it is famous for, and then build the one tool used most in this course: the **moment generating function**, computed by a trick called **completing the square**. At the end we use that trick to derive half of the Black-Scholes formula.

Nothing here needs more than Lessons 1 to 3. Every step is written out; if a line follows from an earlier lesson, the lesson is named.

What you need from earlier lessons

- **Densities** (Lesson 3). A random variable X with density f has $P(a < X < b) = \int_a^b f(x)\,dx$, and the total area $\int f(x)\,dx$ is 1.
- **Expectation of a function** (Lesson 3). $E[g(X)] = \int g(x)f(x)\,dx$. You do not need the distribution of $g(X)$, only the density of X.

- **Linearity and variance** (Lesson 3). $E[aX+b] = aE[X]+b$, $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$ and $\mathrm{Var}(X) = E[X^2] - E[X]^2$.
- **Independence** (Lesson 3). If X and Y are independent, $E[XY] = E[X]E[Y]$. Functions of independent variables are again independent, so also $E[g(X)h(Y)] = E[g(X)]\,E[h(Y)]$.
- **The exponential** (Lesson 1). $e^{a+b} = e^a e^b$, and the Taylor series at 0: every derivative of e^x is e^x, which is 1 at $x = 0$, so

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

- **Substitution and integration by parts** (Lesson 1).

The picture

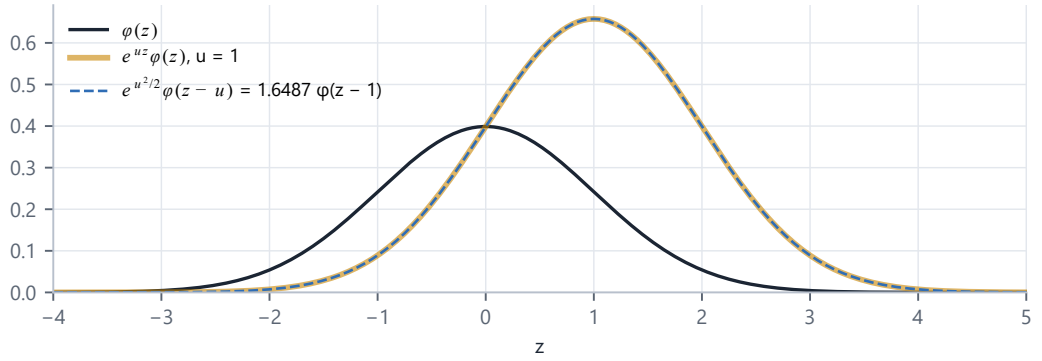


Figure 4.1. Tilting the standard normal density by e^(uz) (u = 1, amber) gives exactly a shifted normal density, scaled by e^(u²/2) (dashed blue). This identity, completing the square, is behind the moment generating function, the Black-Scholes formula and Girsanov's theorem.

Multiply the bell curve by an exponential and you get another bell curve, moved over and scaled up. By the end of this lesson you will be able to prove that, and you will see that almost every closed-form price in this course comes from it.

The standard normal distribution

Definition. A random variable Z has the **standard normal distribution**, written $Z \sim N(0,1)$, if it has the density

$$\varphi(z) = \frac{1}{\sqrt{2\pi}}\, e^{-z^2/2}, \qquad -\infty < z < \infty.$$

(φ is the Greek letter phi.) Read the formula one piece at a time.

- z^2: the height at z and at $-z$ is the same, so the curve is symmetric about 0.
- $e^{-z^2/2}$: this is 1 at $z = 0$ and falls very fast as z moves away. At $z = 1, 2, 3, 4$ it is 0.6065, 0.1353, 0.0111 and 0.000335. Values beyond 3 or 4 are rare: the normal has **thin tails**.
- $1/\sqrt{2\pi}$: a constant chosen only so that the total area is 1. ($\sqrt{2\pi} = 2.5066$.)

Why the constant is $\sqrt{2\pi}$

You can skip this on a first reading, but it is a beautiful argument. It needs Lesson 1 and one new idea: an integral over the whole plane, added up ring by ring. Let

$$I = \int_{-\infty}^{\infty} e^{-x^2/2}\,dx.$$

There is no formula for an antiderivative of $e^{-x^2/2}$, so we cannot compute I directly. The trick is to compute I^2 instead. Writing the second copy of I with the letter y,

$$I^2 = \int e^{-x^2/2}\,dx \int e^{-y^2/2}\,dy = \iint e^{-(x^2+y^2)/2}\,dx\,dy,$$

an integral over the whole (x, y) plane. The height $e^{-(x^2+y^2)/2}$ depends only on the distance $\rho = \sqrt{x^2+y^2}$ from the origin. So cut the plane into thin rings around the origin. The ring at distance ρ with width $d\rho$ has length (circumference) $2\pi\rho$, so its area is about $2\pi\rho\,d\rho$, and on it the height is $e^{-\rho^2/2}$. Adding up the rings:

$$I^2 = \int_0^{\infty} e^{-\rho^2/2}\,2\pi\rho\,d\rho = 2\pi \int_0^{\infty} \rho\, e^{-\rho^2/2}\,d\rho.$$

The last integral is Lesson 1's Example 2: the derivative of $-e^{-\rho^2/2}$ is $\rho\, e^{-\rho^2/2}$, so

$$\int_0^{\infty} \rho\, e^{-\rho^2/2}\,d\rho = \Big[-e^{-\rho^2/2}\Big]_0^{\infty} = 0 - (-1) = 1.$$

So $I^2 = 2\pi$ and $I = \sqrt{2\pi}$. Dividing by $\sqrt{2\pi}$ makes the area exactly 1. □ (The notebook checks it numerically: 1.000000.)

The mean of Z is 0

$$E[Z] = \int_{-\infty}^{\infty} z\,\varphi(z)\,dz = 0.$$

Why. The function $z\varphi(z)$ is **odd**: its value at $-z$ is minus its value at z, because $\varphi(-z) = \varphi(z)$. So the area on the left of 0 is exactly minus the area on the right, and they cancel. (The tails are so thin that both areas are finite, so the cancelling is allowed.)

The variance of Z is 1

We need $E[Z^2] = \int z^2 \varphi(z)\,dz$. The key observation is what happens when we differentiate φ. By the chain rule,

$$\varphi'(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \cdot (-z) = -z\,\varphi(z).$$

So $z^2\varphi(z) = z \cdot z\varphi(z) = -z\,\varphi'(z)$. Integrate by parts, with z as the part we differentiate and $-\varphi'(z)$ as the part we integrate:

$$\int z^2 \varphi(z)\,dz = \Big[-z\,\varphi(z)\Big]_{-\infty}^{\infty} + \int \varphi(z)\,dz.$$

The bracket is 0, because $z\varphi(z) \to 0$ at both ends, and the last integral is the total area, 1. So $E[Z^2] = 1$, and

$$\mathrm{Var}(Z) = E[Z^2] - E[Z]^2 = 1 - 0 = 1.$$

This is why Z is called $N(0,1)$: **mean 0, variance 1**.

The distribution function N

The probability that Z is at most x is the area to the left of x:

$$N(x) = P(Z \le x) = \int_{-\infty}^{x} \varphi(z)\,dz.$$

Like I, this integral has no formula in terms of e^x, logarithms and powers. Computers evaluate it to full precision (in Python, `scipy.stats.norm.cdf(x)`, or `0.5 * (1 + math.erf(x / math.sqrt(2)))`, which is what this course's notebooks use). A few values are worth knowing:

The standard normal distribution function at a few points.

POINT	PROBABILITY BELOW IT
0.000	0.5000
0.500	0.6915
1.000	0.8413
1.645	0.9500
1.960	0.9750
2.000	0.9772
3.000	0.9987

So half the area is below 0, about 84% is below 1, and 97.5% is below 1.96. The numbers 1.645 and 1.96 are the ones behind every "95% interval" you will meet.

Symmetry: $N(-x) = 1 - N(x)$

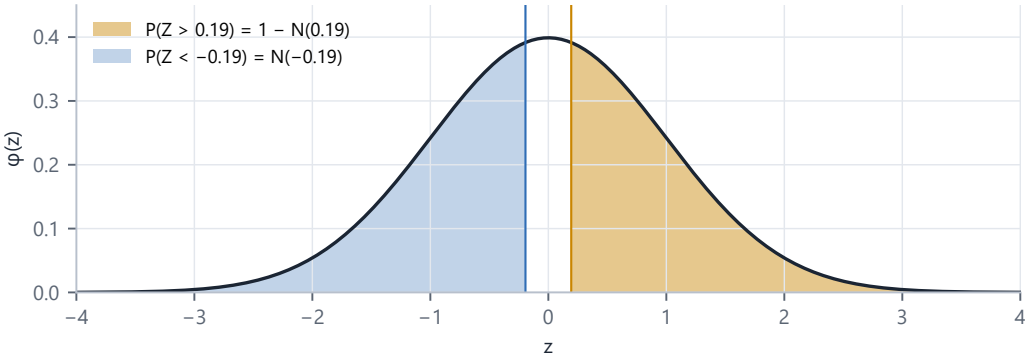


Figure 4.2. The area to the right of 0.19 (amber) equals the area to the left of −0.19 (blue), because the density is symmetric. So 1 − N(x) = N(−x).

Proposition. For every x, $N(-x) = 1 - N(x)$.

Proof. Start from the definition and substitute $w = -z$, so $dz = -dw$; when z runs from $-\infty$ to $-x$, w runs from ∞ down to x:

$$N(-x) = \int_{-\infty}^{-x} \varphi(z)\,dz = \int_{\infty}^{x} \varphi(-w)\,(-dw) = \int_{x}^{\infty} \varphi(w)\,dw.$$

(In the last step we used $\varphi(-w) = \varphi(w)$ and that swapping the limits changes the sign.) The right-hand side is $P(Z > x) = 1 - P(Z \le x) = 1 - N(x)$. $\square$

In words: the area beyond x on the right equals the area beyond $-x$ on the left. You will use this every time you write down a Black-Scholes price.

The general normal, and standardising

Definition. For numbers μ and $s > 0$, the random variable

$$X = \mu + sZ$$

has the **normal distribution with mean μ and variance s^2**, written $X \sim N(\mu, s^2)$.

The names are earned by Lesson 3's rules:

$$E[X] = \mu + sE[Z] = \mu, \qquad \operatorname{Var}(X) = s^2 \operatorname{Var}(Z) = s^2.$$

So the second number in $N(\mu, s^2)$ is the **variance**, and the standard deviation is s.

Its distribution function and density. For any x, because $s > 0$,

$$P(X \le x) = P(\mu + sZ \le x) = P\Big(Z \le \frac{x-\mu}{s}\Big) = N\Big(\frac{x-\mu}{s}\Big).$$

The density is the derivative of the distribution function. By the chain rule (the inner function $(x-\mu)/s$ has derivative $1/s$):

$$f_X(x) = \frac{1}{s}\,\varphi\Big(\frac{x-\mu}{s}\Big) = \frac{1}{s\sqrt{2\pi}}\,e^{-(x-\mu)^2/(2s^2)}.$$

The factor $1/s$ is important: stretching the bell by s makes it s times wider, so it must be s times lower to keep the area 1.

Standardising. The step $P(X \le x) = N\big((x-\mu)/s\big)$ is called standardising: subtract the mean, divide by the standard deviation, and every question about any normal becomes a question about Z.

Worked example: the chance the share ends above the strike

Lesson 5 shows that in the pricing model the share's one-year **log return** $L = \ln(S_T/S_0)$ is normal:

$$L \sim N\big(m, s^2\big), \qquad m = (r - \tfrac{1}{2}\sigma^2)T = 0.0100, \qquad s = \sigma\sqrt{T} = 0.20.$$

(Take this on trust for now; Lesson 5 explains the $-\frac{1}{2}\sigma^2$.) What is $P(S_T > K)$?

Step 1. Turn the event into an event about L. Since $S_T = S_0 e^L$ and the logarithm is increasing,

$$S_T > K \iff e^L > \frac{K}{S_0} \iff L > \ln\frac{K}{S_0} = \ln 1.05 = 0.048790.$$

Step 2. Standardise. Subtract m and divide by s on both sides (dividing by a positive number keeps the inequality):

$$L > 0.0488 \iff \frac{L-m}{s} > \frac{0.0488 - 0.0100}{0.20} = 0.1940.$$

The left side, $(L-m)/s$, is a standard normal Z.

Step 3. Use N and symmetry.

$$P(S_T > K) = P(Z > 0.1940) = 1 - N(0.1940) = N(-0.1940) = 0.4231.$$

Step 4. Check by simulation. Two million simulated log returns give 0.4230.

So the call with strike 105 ends in the money about 42% of the time. Lesson 20 writes this probability as $N(d_-)$; here d_- is -0.1940.

Moments

Definition. The **k-th moment** of a random variable X is $E[X^k]$, for $k = 1, 2, 3, \dots$ (when the expectation is finite).

Each moment tells you something about the shape of the distribution:

- $k = 1$: $E[X]$ is the mean, the centre.
- $k = 2$: together with the mean, $E[X^2]$ gives the variance, $\mathrm{Var}(X) = E[X^2] - E[X]^2$: the spread.
- $k = 3$: measures lopsidedness (**skewness**). A symmetric distribution has $E[(X - \mu)^3] = 0$.
- $k = 4$: measures how heavy the tails are (**kurtosis**): how often very large values occur. In finance, this is the question "how often do crash days happen?"

You can compute each moment as its own integral, $E[Z^k] = \int z^k \varphi(z)\, dz$. For $k = 2$ that took one integration by parts. For $k = 4$ it takes two, for $k = 6$ three, and so on. There is a much better way.

The moment generating function

Definition. The **moment generating function** of a random variable X is

$$M_X(u) = E\left[e^{uX}\right],$$

defined for each real number u at which the expectation is finite.

Read it carefully. The letter u is not random: it is an ordinary number that you choose. For each choice of u, e^{uX} is a random variable and $M_X(u)$ is its average. So M_X is an ordinary function of u, like u^2 or $\sin u$. Using Lesson 3's formulas, it is computed as

$$M_X(u) = \sum_i e^{ux_i}\, p_i \quad \text{(discrete)}, \qquad M_X(u) = \int e^{ux} f(x)\, dx \quad \text{(with a density)}.$$

At $u = 0$ it is always 1, because $e^0 = 1$.

Why it "generates" the moments

Put $x = uX$ into the exponential series:

$$e^{uX} = 1 + uX + \frac{u^2X^2}{2!} + \frac{u^3X^3}{3!} + \dots$$

Take expectations of both sides. By linearity, the expectation of a sum is the sum of the expectations, and u is a constant that comes outside. (With infinitely many terms this needs M_X to be finite for u near 0, which holds for every distribution in this lesson.) So

$$M_X(u) = 1 + E[X]\,u + E[X^2]\,\frac{u^2}{2!} + E[X^3]\,\frac{u^3}{3!} + \cdots = \sum_{k=0}^{\infty} E[X^k]\,\frac{u^k}{k!}.$$

This is the whole idea. The moments are the coefficients of the power series of M_X. There are two ways to read them off.

1. **Differentiate.** Differentiate the series k times and set $u = 0$. Every term with a power of u below k has disappeared, every term above k still contains u and so vanishes at $u = 0$, and the u^k term leaves exactly $E[X^k]$:
$$M_X^{(k)}(0) = E[X^k].$$
2. **Match coefficients.** If you can expand $M_X(u)$ as a power series some other way, the coefficient of u^k is $E[X^k]/k!$.

A first example: a coin

Let $X = 1$ with probability p and $X = 0$ with probability $1 - p$. Then

$$M_X(u) = e^{u\cdot 0}(1-p) + e^{u\cdot 1}p = 1 - p + pe^u.$$

Differentiate: $M_X'(u) = pe^u$ and $M_X''(u) = pe^u$. At $u = 0$:

$$E[X] = M_X'(0) = p, \qquad E[X^2] = M_X''(0) = p, \qquad \mathrm{Var}(X) = p - p^2 = p(1-p).$$

Check directly: $X^2 = X$ (because $0^2 = 0$ and $1^2 = 1$), so $E[X^2] = E[X] = p$. The machine works.

Why we use it

There are four reasons, and the last one is the reason it fills this course.

1. **All moments from one calculation.** Compute one expectation, $E[e^{uX}]$, and then differentiate as often as you like, instead of computing one integral per moment.
2. **Sums become products.** If X and Y are independent, then e^{uX} and e^{uY} are independent too, so
$$M_{X+Y}(u) = E\big[e^{uX}e^{uY}\big] = E\big[e^{uX}\big]E\big[e^{uY}\big] = M_X(u)\,M_Y(u).$$

Finding the distribution of a sum directly is hard; multiplying two functions is easy.

3. **It identifies the distribution.** *Uniqueness theorem:* if two random variables have the same moment generating function on some interval around $u = 0$, they have the same distribution. (The proof needs Fourier or Laplace transforms, beyond this course; we use it as a fact.) So a common strategy is: compute the generating function, recognise it, and conclude what the distribution is.

4. **In finance it is a price formula.** If L is a log return, the share price is $S_T = S_0 e^L$, so

$$E[S_T] = S_0\, E\left[e^L\right] = S_0\, M_L(1), \qquad E\left[S_T^u\right] = S_0^u\, M_L(u).$$

The average share price, the average squared share price, and many prices later in the course, are values of the moment generating function of a log return. Since our log returns are normal, we need the generating function of the normal.

The moment generating function of the standard normal

Proposition. If $Z \sim N(0,1)$, then for every real u,

$$M_Z(u) = E\left[e^{uZ}\right] = e^{u^2/2}.$$

Proof, in four steps.

Step 1. Write it as an integral. By the formula for $E[g(Z)]$ with $g(z) = e^{uz}$, and $e^a e^b = e^{a+b}$:

$$E\left[e^{uZ}\right] = \int_{-\infty}^{\infty} e^{uz}\, \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{uz - z^2/2}\, dz.$$

Step 2. Complete the square in the exponent. We want to write $uz - \frac{1}{2}z^2$ as "minus one half times a square, plus a constant". Take out $-\frac{1}{2}$, then add and subtract u^2 inside the bracket:

$$\begin{aligned} uz - \tfrac{1}{2}z^2 &= -\tfrac{1}{2}\left(z^2 - 2uz\right) \\ &= -\tfrac{1}{2}\left(z^2 - 2uz + u^2 - u^2\right) \\ &= -\tfrac{1}{2}\left((z-u)^2 - u^2\right) \\ &= -\tfrac{1}{2}(z-u)^2 + \tfrac{1}{2}u^2. \end{aligned}$$

Check by multiplying out the last line: $-\frac{1}{2}(z^2 - 2uz + u^2) + \frac{1}{2}u^2 = -\frac{1}{2}z^2 + uz$. Correct.

Step 3. Take the constant outside. Put the completed square back into the integral. The term $\frac{1}{2}u^2$ does not contain z, so $e^{u^2/2}$ comes outside:

$$E\left[e^{uZ}\right] = \frac{1}{\sqrt{2\pi}} \int e^{-\frac{1}{2}(z-u)^2}\, e^{\frac{1}{2}u^2}\, dz = e^{u^2/2} \int \frac{1}{\sqrt{2\pi}} e^{-(z-u)^2/2}\, dz.$$

Step 4. The remaining integral is 1. Substitute $w = z - u$, $dw = dz$; the limits stay $-\infty$ and ∞:

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(z-u)^2/2}\, dz = \int_{-\infty}^{\infty} \varphi(w)\, dw = 1.$$

(Equivalently: the integrand is the density of $N(u, 1)$, and every density has area 1.) So $E[e^{uZ}] = e^{u^2/2}$. $\square$

What Step 2 means in the picture. Steps 2 and 3 say, before integrating,

$$e^{uz}\, \varphi(z) = e^{u^2/2}\, \varphi(z - u).$$

That is Figure 4.1: multiplying the bell curve by e^{uz} gives the same bell curve, moved right by u and scaled up by $e^{u^2/2}$. We will use this identity again below.

The moments of Z

Route 1: differentiate. With $M(u) = e^{u^2/2}$, the chain rule and the product rule give

$$\begin{aligned} M'(u) &= u\, e^{u^2/2}, \\ M''(u) &= (1 + u^2)\, e^{u^2/2}, \\ M'''(u) &= (3u + u^3)\, e^{u^2/2}, \\ M''''(u) &= (3 + 6u^2 + u^4)\, e^{u^2/2}. \end{aligned}$$

(For example, the third line: differentiate $(1 + u^2)e^{u^2/2}$ to get $2u\, e^{u^2/2} + (1 + u^2)\, u\, e^{u^2/2} = (3u + u^3)e^{u^2/2}$.) Now set $u = 0$, where $e^0 = 1$:

$$E[Z] = 0, \qquad E[Z^2] = 1, \qquad E[Z^3] = 0, \qquad E[Z^4] = 3.$$

Route 2: match coefficients. Expand $e^{u^2/2}$ with the exponential series, putting $x = u^2/2$:

$$e^{u^2/2} = 1 + \frac{u^2}{2} + \frac{(u^2/2)^2}{2!} + \frac{(u^2/2)^3}{3!} + \cdots = 1 + \frac{u^2}{2} + \frac{u^4}{8} + \frac{u^6}{48} + \cdots$$

Compare with $M(u) = \sum_k E[Z^k] u^k / k!$, term by term:

- There are no odd powers of u, so **every odd moment is 0** (as symmetry says it must be).
- u^2: $\dfrac{E[Z^2]}{2!} = \dfrac{1}{2}$, so $E[Z^2] = 1$.
- u^4: $\dfrac{E[Z^4]}{4!} = \dfrac{1}{8}$, so $E[Z^4] = \dfrac{24}{8} = 3$.
- In general, the coefficient of u^{2j} is $\dfrac{1}{2^j\, j!}$, so

$$E[Z^{2j}] = \frac{(2j)!}{2^j\, j!} = 1 \times 3 \times 5 \times \cdots \times (2j-1).$$

Check by simulation. Two million standard normal draws give $E[Z] \approx 0.0001$, $E[Z^2] \approx 0.9997$, $E[Z^3] \approx 0.0017$, $E[Z^4] \approx 2.9972$ and $E[Z^6] \approx 14.976$, against 0, 1, 0, 3 and 15.

Remember $E[Z^4] = 3$. Lesson 6 and Lesson 14 use it to show that the squared steps of a random path add up to time, which is the heart of Itô's lemma.

The moment generating function of any normal

Proposition. If $X \sim N(\mu, s^2)$, then

$$M_X(u) = E\big[e^{uX}\big] = e^{u\mu + \frac{1}{2}u^2 s^2}.$$

Proof. Write $X = \mu + sZ$. Then $e^{uX} = e^{u\mu}\, e^{(us)Z}$, and $e^{u\mu}$ is a constant, so it comes outside the expectation:

$$E\big[e^{uX}\big] = e^{u\mu}\, E\big[e^{(us)Z}\big] = e^{u\mu}\, M_Z(us) = e^{u\mu}\, e^{(us)^2/2} = e^{u\mu + \frac{1}{2}u^2 s^2}. \quad \square$$

The only trick is to see that $E[e^{(us)Z}]$ is M_Z evaluated at the number us. (Exercise 4.6 asks you to get the same answer straight from the density.)

The case $u = 1$ is the most-used formula in this course. For any normal X,

$$E\big[e^{X}\big] = e^{\,\text{mean} + \frac{1}{2}\text{variance}}.$$

The exponential of a normal

Why the average is not e^{mean}

Take $X = \sigma Z$ with $\sigma = 20\%$. Its mean is 0. A tempting (and wrong) shortcut is

$$E\big[e^{\sigma Z}\big] \overset{?}{=} e^{E[\sigma Z]} = e^0 = 1.$$

The formula above gives instead $E[e^{\sigma Z}] = e^{\sigma^2/2} = 1.020201$, and two million simulated draws agree: 1.020219. Where does the extra 2% come from?

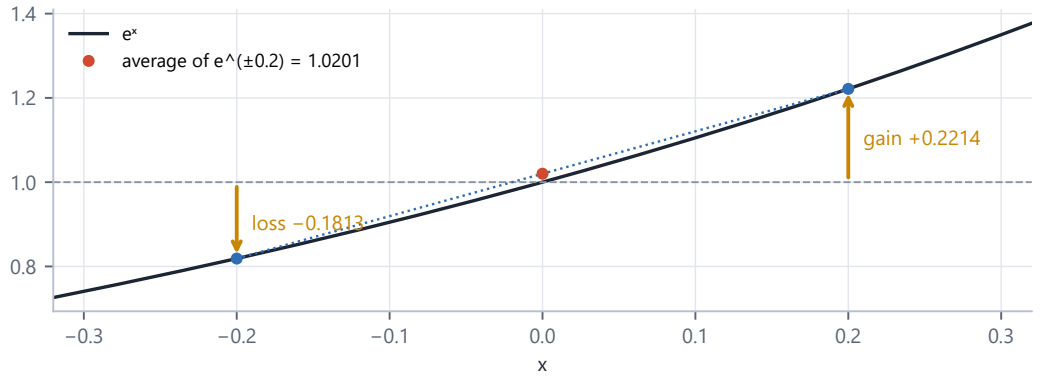


Figure 4.3. The curve e^x bends upwards. A move of +0.2 gains more (amber, up) than a move of −0.2 loses (amber, down), so the average of the two outcomes (red dot) sits above $e^0 = 1$.

You can see it without any integral. Replace σZ by a simpler variable with the same mean and variance: $+0.2$ or -0.2, with probability one half each. Then

$$E\left[e^{X}\right] = \frac{e^{0.2} + e^{-0.2}}{2} = 1.020067.$$

The up move multiplies by $e^{0.2}$, a gain of 0.2214, but the down move multiplies by $e^{-0.2}$, a loss of only 0.1813. Because the curve e^x bends upwards (it is **convex**), equal-sized moves up and down do not cancel: the gains win. Lesson 5 turns this into a general rule, Jensen's inequality.

The average share price

In the pricing model $S_T = S_0 e^{L}$ with $L \sim N\big((r - \tfrac{1}{2}\sigma^2)T,\ \sigma^2 T\big)$. With the $u = 1$ formula:

$$E[S_T] = S_0\, E\left[e^{L}\right] = S_0 \exp\Big(\underbrace{(r - \tfrac{1}{2}\sigma^2)T}_{\text{mean}} + \tfrac{1}{2}\ \underbrace{\sigma^2 T}_{\text{variance}} \Big)$$
$$= S_0\, e^{rT}.$$

The $-\frac{1}{2}\sigma^2 T$ in the mean exactly cancels the $+\frac{1}{2}\sigma^2 T$ that the exponential adds. On average, the share grows at the risk-free rate, like money in the bank, which is what a pricing model must say. With the course's numbers, $100e^{0.03} = 103.0455$; two million simulated prices average 103.0473. Had we wrongly used e^{mean}, we would have got $100e^{0.01} = 101.0050$. Lesson 5 takes this further.

Sums of independent normals

Proposition. If $X_1 \sim N(\mu_1, s_1^2)$ and $X_2 \sim N(\mu_2, s_2^2)$ are independent, then

$$X_1 + X_2 \sim N\big(\mu_1 + \mu_2,\ s_1^2 + s_2^2\big).$$

Proof, using reasons 2 and 3 for the generating function.

Step 1. Each has a known generating function: $M_{X_1}(u) = e^{u\mu_1 + \frac{1}{2}u^2 s_1^2}$ and $M_{X_2}(u) = e^{u\mu_2 + \frac{1}{2}u^2 s_2^2}$.

Step 2. By independence, the generating function of the sum is the product. Multiplying exponentials adds the exponents:

$$M_{X_1+X_2}(u) = e^{u\mu_1 + \frac{1}{2}u^2 s_1^2}\, e^{u\mu_2 + \frac{1}{2}u^2 s_2^2} = e^{u(\mu_1+\mu_2) + \frac{1}{2}u^2 (s_1^2 + s_2^2)}.$$

Step 3. This is exactly the generating function of $N(\mu_1 + \mu_2,\ s_1^2 + s_2^2)$. By the uniqueness theorem, that is the distribution of $X_1 + X_2$. □

Notice what is new. Lesson 3 already showed that means add and, for independent variables, variances add, for *any* distribution. What the proposition adds is that the sum is still **normal**. Normals are closed under adding independent copies; that is why a path made of many small normal steps stays normal at every time.

Example: from days to a year. Split the year into 252 trading days, and let each day's log return be independent and normal with

$$\text{mean } \frac{0.01}{252} = 4.0e-05, \qquad \text{variance } \frac{0.2^2}{252} = 1.6e-04.$$

By the proposition (applied 251 times), the year's log return is normal with mean $252 \times 0.01/252 = 0.01$ and variance $252 \times 0.2^2/252 = 0.2^2$: exactly the yearly distribution we started from. The daily **standard deviation** is $0.2/\sqrt{252} = 0.01260$, not $0.2/252$. Simulated over 100,000 years: mean 0.00998, standard deviation 0.19982.

Standard deviations do not add; variances do. That is why volatility grows with $\sqrt{\text{time}}$.

Completing the square: the general tool

Option payoffs are paid only on some outcomes: a call pays only when $S_T > K$. To handle "only on some outcomes" we use an **indicator**:

$$\mathbf{1}\{Z > a\} = \begin{cases} 1 & \text{if } Z > a, \\ 0 & \text{otherwise.} \end{cases}$$

Multiplying by it switches off the outcomes we do not want, so $E\big[g(Z)\,\mathbf{1}\{Z > a\}\big] = \int_a^\infty g(z)\,\varphi(z)\,dz$: the integral only runs over $z > a$.

Proposition. For any numbers u and a,

$$E\big[e^{uZ}\,\mathbf{1}\{Z>a\}\big] = e^{u^2/2}\,N(u-a).$$

Proof.

Step 1. Write it as an integral over $z>a$, and use the completed square $e^{uz}\varphi(z) = e^{u^2/2}\varphi(z-u)$ from the generating-function proof:

$$E\big[e^{uZ}\,\mathbf{1}\{Z>a\}\big] = \int_a^\infty e^{uz}\varphi(z)\,dz = e^{u^2/2}\int_a^\infty \varphi(z-u)\,dz.$$

Step 2. Substitute $w = z-u$, $dw = dz$. **The limits move too**: when $z=a$, $w=a-u$; when $z\to\infty$, $w\to\infty$:

$$e^{u^2/2}\int_{a-u}^\infty \varphi(w)\,dw = e^{u^2/2}\,P(Z>a-u).$$

Step 3. By symmetry, $P(Z>a-u) = 1-N(a-u) = N(u-a)$. □

Check. Let $a\to-\infty$: the indicator is then always 1, and $N(u-a)\to 1$, so the formula becomes $E[e^{uZ}] = e^{u^2/2}$, the generating function again.

The first term of the Black-Scholes formula

Lesson 20 shows that a call's price is

$$e^{-rT}E\big[\max(S_T-K,0)\big] = \underbrace{e^{-rT}E\big[S_T\,\mathbf{1}\{S_T>K\}\big]}_{\text{this section}} - Ke^{-rT}P(S_T>K).$$

We already found $P(S_T>K)$ above. Here is the first term, one step at a time.

Step 1. Write S_T with a standard normal.

$$S_T = S_0\,\exp\Big((r-\tfrac12\sigma^2)T + \sigma\sqrt{T}\,Z\Big).$$

Step 2. Turn the event $S_T>K$ into an event about Z. Take logarithms, then solve for Z:

$$\begin{aligned} S_T>K &\iff (r-\tfrac12\sigma^2)T + \sigma\sqrt{T}\,Z > \ln\frac{K}{S_0} \\ &\iff Z > \frac{\ln(K/S_0)-(r-\tfrac12\sigma^2)T}{\sigma\sqrt{T}} = -d_-, \end{aligned}$$

where, using $\ln(K/S_0) = -\ln(S_0/K)$,

$$d_- = \frac{\ln(S_0/K)+(r-\tfrac12\sigma^2)T}{\sigma\sqrt{T}}.$$

Step 3. Take the constants outside. Discounting and the drift combine:

$$e^{-rT} S_T = S_0\, e^{-rT} e^{(r-\frac{1}{2}\sigma^2)T} e^{\sigma\sqrt{T}Z} = S_0\, e^{-\frac{1}{2}\sigma^2 T}\, e^{\sigma\sqrt{T}Z}.$$

So

$$e^{-rT} E\big[S_T\, \mathbf{1}\{S_T > K\}\big] = S_0\, e^{-\frac{1}{2}\sigma^2 T}\, E\big[e^{\sigma\sqrt{T}Z}\, \mathbf{1}\{Z > -d_-\}\big].$$

Step 4. Use the proposition with $u = \sigma\sqrt{T}$ and $a = -d_-$, so $u - a = \sigma\sqrt{T} + d_-$:

$$E\big[e^{\sigma\sqrt{T}Z}\, \mathbf{1}\{Z > -d_-\}\big] = e^{\frac{1}{2}\sigma^2 T}\, N\big(d_- + \sigma\sqrt{T}\big).$$

Step 5. Simplify. The factors $e^{-\frac{1}{2}\sigma^2 T}$ and $e^{\frac{1}{2}\sigma^2 T}$ cancel:

$$e^{-rT} E\big[S_T\, \mathbf{1}\{S_T > K\}\big] = S_0\, N(d_+),$$

where

$$d_+ = d_- + \sigma\sqrt{T} = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}.$$

Numbers. d_- = -0.193951 (minus the 0.1940 of the worked example), $d_+ = 0.006049$, $N(d_+) =$ 0.502413, so $S_0 N(d_+) =$ 50.2413. Two million simulated prices give 50.2316.

The completed square as a price: $S_0N(d_+)$ against simulation.

```
d_minus = (math.log(S0 / K) + (r - 0.5 * sigma**2) * T) / (sigma * math.sqrt(T))
d_plus = (math.log(S0 / K) + (r + 0.5 * sigma**2) * T) / (sigma * math.sqrt(T))
term = math.exp(-r * T) * float((ST * (ST > K)).mean())
```

Every closed-form option price in this course is built this way: write the price with Z, turn the exercise event into $Z > a$, and complete the square.

The average size of a normal: $E|Z|$

$E[Z] = 0$ because ups and downs cancel. If we ignore the direction and ask for the average *size* of Z, we need $E|Z|$. Lesson 6 uses it to measure the length of a random path.

Proposition. $E|Z| = \sqrt{2/\pi}$.

Proof.

Step 1. Use symmetry. $|z|\varphi(z)$ is the same at z and $-z$, so the area on the left of 0 equals the area on the right:

$$E|Z| = \int_{-\infty}^{\infty} |z|\, \varphi(z)\, dz = 2\int_0^{\infty} z\, \varphi(z)\, dz = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} z\, e^{-z^2/2}\, dz.$$

Step 2. Substitute $v = z^2/2$, so $dv = z\, dz$; v runs from 0 to ∞:

$$\int_0^\infty z\, e^{-z^2/2}\, dz = \int_0^\infty e^{-v}\, dv = 1.$$

Step 3. So $E|Z| = 2/\sqrt{2\pi} = \sqrt{2/\pi}$. $\square$

That is 0.797885; simulation gives 0.797754. Notice that it is less than $\sqrt{E[Z^2]} = 1$: the typical size of a normal step is a bit smaller than its standard deviation.

What to remember

- $Z \sim N(0,1)$ has density $\varphi(z) = e^{-z^2/2}/\sqrt{2\pi}$, mean 0, variance 1. $X = \mu + sZ \sim N(\mu, s^2)$.
- Standardise: $P(X \le x) = N\big((x-\mu)/s\big)$. Symmetry: $N(-x) = 1 - N(x)$.
- $M_X(u) = E[e^{uX}]$; its power-series coefficients are $E[X^k]/k!$, and $M_X^{(k)}(0) = E[X^k]$.
- Independent sums: $M_{X+Y} = M_X M_Y$. Same generating function means same distribution.
- $E[e^{uZ}] = e^{u^2/2}$, and for any normal, $E[e^X] = e^{\text{mean} + \frac{1}{2}\text{variance}}$.
- $E[Z^2] = 1$, $E[Z^4] = 3$, odd moments 0, $E|Z| = \sqrt{2/\pi}$.
- Completing the square: $e^{uz}\varphi(z) = e^{u^2/2}\varphi(z-u)$, so $E[e^{uZ}\mathbf{1}\{Z > a\}] = e^{u^2/2}N(u-a)$.
- $e^{-rT}E[S_T \mathbf{1}\{S_T > K\}] = S_0 N(d_+)$.

Common mistakes

- **"If $Z \sim N(0,1)$ then $\mu + sZ \sim N(\mu, s)$."** The second number is the *variance*: $N(\mu, s^2)$. Writing standard deviations where variances belong is the most common slip in this course.
- **Forgetting the $1/s$ in the density.** The density of $N(\mu, s^2)$ is $\frac{1}{s}\varphi\big((x-\mu)/s\big)$, not $\varphi\big((x-\mu)/s\big)$. Without the $1/s$ the area is s, not 1.
- **"Standard deviations add."** For independent normals the *variances* add. Twelve independent monthly moves of size s make a yearly move of size $s\sqrt{12}$, not $12s$.
- **"$E[e^{\sigma Z}] = e^0 = 1$ because $E[Z] = 0$."** The exponential is convex; the average of $e^{\sigma Z}$ is $e^{\sigma^2/2}$, not $e^{E[\sigma Z]}$.
- **Forgetting to move the limits in a substitution.** In $\int_a^\infty \varphi(z-u)\, dz$, putting $w = z - u$ moves the lower limit to $a - u$. Leaving it at a gives the wrong answer.
- **Treating u as random.** In $M_X(u)$, u is a fixed number you choose; the generating function is an ordinary function of u.

Exercises

Solutions are in *Part 1: Solutions*.

4.1 Let $X \sim N(100, 20^2)$. Compute $P(X > 105)$ and $P(90 < X < 110)$ using N and its symmetry.

4.2 Use the moment generating function $M(u) = e^{u^2/2}$ to find $E[Z^6]$. (Expand M as a power series and match coefficients.)

4.3 Use moment generating functions to show that if $X_1, \dots, X_n$ are independent $N(\mu, s^2)$, then $X_1 + \cdots + X_n \sim N(n\mu, ns^2)$.

4.4 Show by completing the square that $E[e^{uZ}\mathbf{1}\{Z > a\}] = e^{u^2/2}N(u - a)$. Evaluate it for $u = 0.2$, $a = 0$.

4.5 Let $X = +1$ or -1 with probability one half each. Find its moment generating function, and use it to find $E[X^2]$ and $E[X^4]$. Compare $E[X^4]$ with $E[Z^4] = 3$: what does the difference say about the tails?

4.6 Find the moment generating function of $X \sim N(\mu, s^2)$ directly from its density, by completing the square, without writing $X = \mu + sZ$.

4.7 *(Code)* Estimate $E[Z^4]$ and $E|Z|$ by simulation with 10^3, 10^5 and 10^7 samples. Which settles faster, and why?

4.8 *(Code)* Check $e^{-rT}E[S_T\mathbf{1}\{S_T > K\}] = S_0 N(d_+)$ by simulation with the course's numbers.

Self-test

1. What is $\mathrm{Var}(Z^2)$?
2. What is $E[e^{-0.5Z}]$?
3. Why is $N(-x) = 1 - N(x)$, and where did this lesson use it?

Companion notebook: `s5e04_normal_distribution.ipynb`. Every number in this chapter is computed by it.

LESSON 5 · VIDEO 5

Lognormal Prices and Jensen's Inequality

Why price models are written with normal log returns; the lognormal distribution, its density, mean, variance, median and mode, each derived step by step; convex functions and Jensen's inequality, and how they explain the $-\sigma^2/2$ in the share price model; and the central limit theorem, which explains why log returns look normal at all.

Why this lesson

The Black-Scholes model says a share's log return over T years is normal:

$$\ln \frac{S_T}{S_0} = (r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z, \qquad Z \sim N(0,1).$$

Lesson 4 used this formula and promised to explain it. Three questions follow, and this lesson answers them:

1. **What does it make the price itself?** The price is **lognormal**. We derive its density, mean, variance, median and mode.
2. **Where does the odd-looking $-\frac{1}{2}\sigma^2$ come from?** From **Jensen's inequality**, a fact about averaging curved functions. It is also the reason options have value.
3. **Why should log returns be normal in the first place?** Because of the **central limit theorem**: a sum of many small independent pieces is close to normal, whatever the pieces look like.

What you need from earlier lessons

- **Log returns** (Lesson 1). $\ln(S_1/S_0)$; log returns over consecutive periods add up.
- **Convexity** (Lesson 1). $f'' > 0$ means the curve bends upwards. Taylor's theorem with remainder.
- **Expectation, variance, median ideas** (Lesson 3). $E[g(X)] = \int gf$, $\mathrm{Var}(X) = E[X^2] - E[X]^2$, densities and distribution functions, $F' = f$.

- **The normal distribution** (Lesson 4). Standardising, $N(-x) = 1 - N(x)$, the moment generating function $E[e^{uZ}] = e^{u^2/2}$, and $E[e^X] = e^{\text{mean} + \frac{1}{2}\text{variance}}$ for a normal X.

The picture

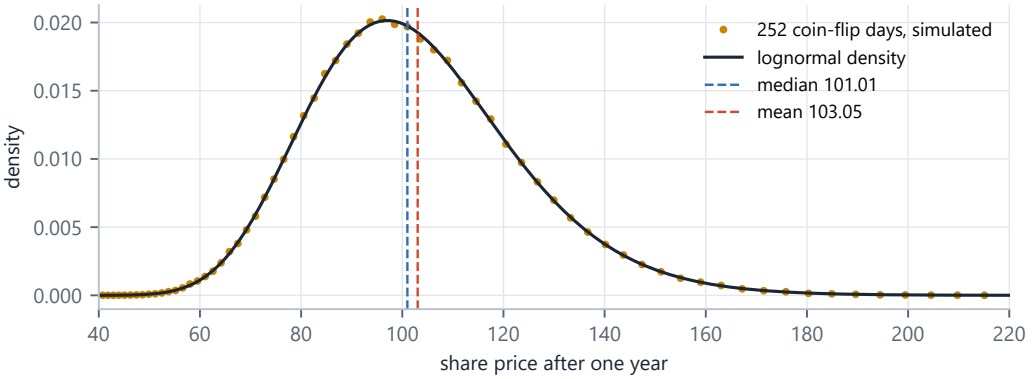


Figure 5.1. Year-end prices from a model in which each of 252 days is a coin flip, up or down 1.26% (dots), against the lognormal density (line). The mean sits above the median: a few big winners pull the average up.

Two things to notice. First, even though each day is just a coin flip, the year-end prices follow the smooth lognormal curve almost exactly: that is the central limit theorem. Second, the curve **leans**: it has a long tail to the right, and its mean (red) sits above its median (blue). Both facts are explained in this lesson.

From log returns to prices

Why model log returns, not prices

Why not simply say the price itself is normal? Two reasons:

- **A normal can be negative**, and a share price cannot. An exponential, $S_0 e^L$, is always positive whatever L is.
- **Log returns add up over time** (Lesson 1), so a year's log return is a sum of daily ones. Sums of independent normals are normal (Lesson 4), so a model with normal daily log returns is consistent: the weekly, monthly and yearly log returns are normal too.

The lognormal distribution

Definition. A positive random variable X is **lognormal** if $\ln X$ is normal.

In our model $L = \ln(S_T/S_0)$ is normal with

$$\text{mean } m = (r - \tfrac{1}{2}\sigma^2)T, \qquad \text{variance } v^2 = \sigma^2 T,$$

so $\ln S_T = \ln S_0 + L$ is normal too, and $S_T = S_0 e^L$ is lognormal. With the course's numbers, $m = 0.0100$ and $v = 0.2$.

Its density

Step 1. The distribution function. Since ln is increasing, $S_T \le s$ exactly when $L \le \ln(s/S_0)$. Standardise (Lesson 4):

$$F(s) = P(S_T \le s) = P\Big(L \le \ln\frac{s}{S_0}\Big) = N\Big(\frac{\ln(s/S_0) - m}{v}\Big), \qquad s > 0.$$

Step 2. Differentiate. The density is F'. By the chain rule, with outer function N (derivative φ) and inner function $\big(\ln(s/S_0) - m\big)/v$ (derivative $\frac{1}{vs}$, because $\frac{d}{ds}\ln s = 1/s$):

$$f(s) = \varphi\Big(\frac{\ln(s/S_0) - m}{v}\Big) \cdot \frac{1}{v\,s}.$$

Step 3. Write it out.

$$f(s) = \frac{1}{s\,v\sqrt{2\pi}} \exp\Big(-\frac{\big(\ln(s/S_0) - m\big)^2}{2v^2}\Big), \qquad s > 0.$$

It is a normal bell curve in the variable $\ln s$, with an extra factor $1/s$. That factor makes the density higher for small s and lower for large s, which pushes the peak to the left and stretches the tail to the right: the lean you see in the picture.

The mean and variance of the price

Proposition. For the lognormal share,

$$E[S_T] = S_0 e^{rT}, \qquad E[S_T^2] = S_0^2 e^{(2r+\sigma^2)T}, \qquad \operatorname{Var}(S_T) = S_0^2 e^{2rT}\big(e^{\sigma^2 T} - 1\big).$$

Proof. Write $S_T = S_0 e^{m + vZ}$ and use $E[e^{uZ}] = e^{u^2/2}$ from Lesson 4.

Step 1. The mean ($u = v$). The constant $S_0 e^m$ comes outside:

$$E[S_T] = S_0 e^m\, E[e^{vZ}] = S_0 e^{m + \frac{1}{2}v^2} = S_0 \exp\Big((r - \tfrac{1}{2}\sigma^2)T + \tfrac{1}{2}\sigma^2 T\Big) = S_0 e^{rT}.$$

Step 2. The second moment ($u = 2v$). Square first: $S_T^2 = S_0^2 e^{2m + 2vZ}$, so

$$E[S_T^2] = S_0^2 e^{2m}\, E[e^{2vZ}] = S_0^2 e^{2m + 2v^2} = S_0^2 \exp\Big((2r - \sigma^2)T + 2\sigma^2 T\Big) = S_0^2 e^{(2r+\sigma^2)T}.$$

Step 3. The variance. $E[S_T]^2 = S_0^2 e^{2rT}$, so

$$\operatorname{Var}(S_T) = S_0^2 e^{(2r+\sigma^2)T} - S_0^2 e^{2rT} = S_0^2 e^{2rT}\left(e^{\sigma^2 T} - 1\right). \square$$

Numbers. $E[S_T] = 103.0455$, $E[S_T^2] = 11051.71$ (the $m(1)$ of Lesson 2's differential equation), $\operatorname{Var}(S_T) = 433.34$ and the standard deviation is 20.8169. Two million simulated prices give mean 103.0473 and standard deviation 20.8162.

Median and mode

Definitions. The **median** is the value M with $P(X \le M) = \frac{1}{2}$: half the outcomes are below it, half above. The **mode** is where the density is highest: the single most likely region.

The median. Since ln is increasing, $S_T \le M$ exactly when $\ln S_T \le \ln M$. So the median of S_T is e to the power of the median of $\ln S_T$. A normal is symmetric, so its median is its mean, $\ln S_0 + m$. Hence

$$\operatorname{median}(S_T) = S_0 e^{m} = S_0 e^{(r-\frac{1}{2}\sigma^2)T} = 101.0050$$

(simulated: 101.0051).

The mode. Setting the derivative of the density to zero (Exercise 5.4) gives $S_0 e^{(r-\frac{3}{2}\sigma^2)T} = 97.0446$.

Mode < median < mean. The three numbers are 97.04, 101.01 and 103.05. The median is below the mean by the factor $e^{-\frac{1}{2}\sigma^2 T} = 0.9802$. The reason is the right-hand tail: a price can rise without limit but cannot fall below 0, so a few large rises pull the average up, above the typical outcome. Only 46% of simulated prices end above the mean.

Convex functions

Lesson 1 called a function **convex** when $f'' \ge 0$: it bends upwards. Here are three equivalent ways to say it, each useful:

1. **Bending.** $f''(x) \ge 0$ everywhere.
2. **Tangent lines lie below.** At every point m, the curve lies above its tangent line: $f(x) \ge f(m) + f'(m)(x - m)$ for all x.
3. **Chords lie above.** The straight line joining two points of the curve lies above the curve between them.

Why 1 gives 2. Taylor's theorem with remainder (Lesson 1) says

$$f(x) = f(m) + f'(m)(x-m) + \tfrac{1}{2} f''(\xi)(x-m)^2$$

for some ξ between m and x. If $f'' \geq 0$, the last term is ≥ 0, so $f(x) \geq f(m) + f'(m)(x-m)$.

Examples. Convex: e^x, x^2, and the call payoff $\max(x-K, 0)$ (a kink, but it bends the right way). **Concave** (bending downwards, $f'' \leq 0$, all inequalities reversed): $\ln x$, $\sqrt{x}$.

Jensen's inequality

Theorem (Jensen). If f is convex, then

$$E[f(X)] \geq f(E[X]).$$

The inequality is strict unless X is constant or f is a straight line where X lives. For concave f it goes the other way: $E[f(X)] \leq f(E[X])$.

Proof, in three steps.

Step 1. Let $m = E[X]$ (a number). By property 2 of convex functions, for every possible value x,

$$f(x) \geq f(m) + c\,(x - m),$$

where c is the slope of the tangent at m. (For a function with a kink, like the call payoff, any line through $(m, f(m))$ that stays below the curve will do.)

Step 2. The inequality holds for every value, so it holds for the random variable: $f(X) \geq f(m) + c\,(X - m)$. Take expectations of both sides (averaging preserves $\geq$) and use linearity:

$$E[f(X)] \geq f(m) + c\,\big(E[X] - m\big).$$

Step 3. $E[X] - m = 0$, so $E[f(X)] \geq f(m) = f(E[X])$. $\square$

The picture behind it. Lesson 4's two-point example: if X is $+0.2$ or -0.2 with equal probability, $E[e^X] = (e^{0.2} + e^{-0.2})/2 = 1.0201 > e^0 = 1$. The average of two points on a convex curve is the midpoint of the chord, which lies above the curve.

Where the $-\frac{1}{2}\sigma^2$ comes from

Deriving it

Part 2 will show that, for pricing, the share must grow **on average** at the risk-free rate: $E[S_T] = S_0 e^{rT}$. Suppose the log return is normal with variance $\sigma^2 T$ and an unknown mean μT. What must μ be?

Step 1. By $E[e^X] = e^{\text{mean} + \frac{1}{2}\text{variance}}$ (Lesson 4):

$$E[S_T] = S_0\, E[e^L] = S_0\, e^{\mu T + \frac{1}{2}\sigma^2 T}.$$

Step 2. Set this equal to $S_0 e^{rT}$ and compare the exponents:

$$\mu T + \tfrac{1}{2}\sigma^2 T = rT \quad \Longrightarrow \quad \mu = r - \tfrac{1}{2}\sigma^2.$$

That is the whole story. The exponential is convex, so by Jensen $E[e^L] > e^{E[L]}$: the average of the exponential is pushed up by exactly $e^{\frac{1}{2}\sigma^2 T}$. To keep the average price growing at r, the log return's mean must be pulled down by exactly $\frac{1}{2}\sigma^2$. Part 3 derives the same term again, from Itô's lemma.

What goes wrong without it

If we wrote $S_T = S_0 e^{rT + \sigma\sqrt{T}Z}$, without the correction, the average price would be $S_0 e^{rT} e^{\frac{1}{2}\sigma^2 T}$: too big by the factor 1.020201. In simulation, using the same random numbers, the average comes out at 105.1290 instead of 103.0455.

The same random numbers, with and without the $-\sigma^2/2$.

```
naive = S0 * np.exp(r * T + sigma * math.sqrt(T) * Z)
call_avg = float(np.maximum(ST - K, 0).mean())
```

Concave functions: average growth and typical growth

The logarithm is concave, so Jensen goes the other way: $E[\ln S_T] \le \ln E[S_T]$. In growth-rate terms:

- the **average log return** is $E[\ln(S_T/S_0)] = (r - \frac{1}{2}\sigma^2)T = 0.0100$ (simulated: 0.0100);
- the **log of the average return** is $\ln\big(E[S_T]/S_0\big) = rT = 0.0300$.

The typical (median) path grows at $r - \frac{1}{2}\sigma^2$, slower than the average at r. The gap, $\frac{1}{2}\sigma^2 = 0.02$ a year here, is often called **volatility drag**: the more volatile the share, the further its typical outcome falls behind its average one.

Options are convex too

The call payoff $\max(s - K, 0)$ is convex, so Jensen gives

$$E[\max(S_T - K, 0)] \ge \max(E[S_T] - K, 0).$$

Here the right-hand side is 0.00, because the average price, 103.05, is below the strike. But the left-hand side is 7.3455. An option is worth more than its payoff at the average price **because** its payoff is convex: it gains from the ups and is protected from the

downs. Volatility increases the gap, which is why options are worth more when volatility is higher.

The central limit theorem

The statement

Theorem (central limit theorem). Let $Y_1, Y_2, \ldots$ be independent and identically distributed, with mean μ and variance $v^2 < \infty$. Then the standardised sum

$$\frac{Y_1 + \cdots + Y_n - n\mu}{v\sqrt{n}}$$

is approximately $N(0,1)$ for large n. Precisely: for every interval, the probability that the standardised sum lands in it tends to the standard normal probability of that interval ("convergence in distribution").

Read the standardisation carefully: the sum has mean $n\mu$ and variance nv^2 (Lesson 3, independent variances add), so subtracting $n\mu$ and dividing by $v\sqrt{n}$ gives mean 0 and variance 1. The theorem says that, beyond mean 0 and variance 1, the **shape** becomes normal, whatever the shape of the individual Y_i.

Why it is true (a sketch with generating functions)

Step 1. Standardise each piece. Let $X_i = (Y_i - \mu)/v$, which has mean 0 and variance 1, so $E[X_i^2] = 1$. The standardised sum is $(X_1 + \cdots + X_n)/\sqrt{n}$.

Step 2. Its generating function. By independence (Lesson 4, sums become products),

$$E\left[e^{u(X_1+\cdots+X_n)/\sqrt{n}}\right] = \left(M_X\left(u/\sqrt{n}\right)\right)^n.$$

Step 3. Taylor. Expand $M_X(w) = 1 + E[X]\,w + E[X^2]\,\frac{w^2}{2} + \cdots = 1 + \frac{w^2}{2} + \ldots$, since $E[X] = 0$ and $E[X^2] = 1$. With $w = u/\sqrt{n}$:

$$M_X\left(u/\sqrt{n}\right) = 1 + \frac{u^2}{2n} + (\text{smaller terms}).$$

Step 4. The limit. Exercise 1.4 showed $(1 + a/n)^n \to e^a$. With $a = u^2/2$:

$$\left(1 + \frac{u^2}{2n} + \ldots\right)^n \longrightarrow e^{u^2/2},$$

the generating function of $N(0,1)$. By uniqueness (Lesson 4), the limit is standard normal. □

(This sketch assumes the generating function exists. A full proof uses a close relative, the characteristic function $E[e^{iuX}]$, which always exists.)

Why log returns look normal

A year's log return is the sum of 252 daily log returns. Make each day as un-normal as possible: a coin flip, up or down by $\sigma\sqrt{\Delta t} = 1.26\%$ around the drift. A single day is nothing like a bell curve; it has just two values. The year still is, almost:

Quantiles of the year-end price: 400,000 simulated years of coin-flip days, against the lognormal.

QUANTILE	COIN-FLIP DAYS	LOGNORMAL
0.01	64.17	63.43
0.10	78.51	78.17
0.50	101.01	101.01
0.90	129.95	130.51
0.99	158.97	160.84

The median matches exactly and the 1% and 99% points within about 1%. Real daily returns are not coin flips either, but the same theorem pulls their sum towards normal.

A caveat. It is not the whole story. Real returns have fatter tails than the normal (big crashes are more common than the model says), and their volatility changes over time. Those are reasons for the "volatility smile" you will meet in options markets. The lognormal model is the starting point, not the last word.

What to remember

- $S_T = S_0 e^L$ with $L \sim N\big((r - \tfrac{1}{2}\sigma^2)T, \sigma^2 T\big)$ is lognormal: positive, skewed right.
- Density: $f(s) = \frac{1}{s\,v}\varphi\big((\ln(s/S_0) - m)/v\big)$.
- $E[S_T] = S_0 e^{rT}$, $E[S_T^2] = S_0^2 e^{(2r+\sigma^2)T}$, $\mathrm{Var}(S_T) = S_0^2 e^{2rT}(e^{\sigma^2 T} - 1)$.
- Median $S_0 e^{(r-\frac{1}{2}\sigma^2)T}$, mode $S_0 e^{(r-\frac{3}{2}\sigma^2)T}$: mode < median < mean.
- Jensen: convex $f \Rightarrow E[f(X)] \geq f(E[X])$; concave the other way.
- The $-\frac{1}{2}\sigma^2$ is exactly what makes $E[S_T] = S_0 e^{rT}$.
- Central limit theorem: standardised sums of independent, identically distributed pieces become $N(0,1)$.

Common mistakes

- **"The median price is $S_0 e^{rT}$."** That is the mean. The median is lower, $S_0 e^{(r-\frac{1}{2}\sigma^2)T}$, and more than half the paths end below the mean.
- **"The $-\frac{1}{2}\sigma^2$ is a fudge."** It is Jensen's gap, exactly: without it the average price would grow faster than r.
- **Forgetting the $1/s$ in the lognormal density.** It comes from the chain rule ($\frac{d}{ds}\ln s = 1/s$), and without it the area is not 1.
- **"The central limit theorem says prices are normal."** It says the *sum of log returns* is close to normal; prices, which multiply, are then close to lognormal.
- **Applying Jensen in the wrong direction.** Convex (bends up, like e^x): average of $f \geq f$ of average. Concave (bends down, like ln): $\leq$.

Exercises

Solutions are in *Part 1: Solutions*.

5.1 Use $E[e^{uZ}] = e^{u^2/2}$ to derive $E[S_T]$, $E[S_T^2]$ and $\mathrm{Var}(S_T)$ for the lognormal share. Evaluate the standard deviation with the course's numbers.

5.2 Show that the median of S_T is $S_0 e^{(r-\frac{1}{2}\sigma^2)T}$. Why is it below the mean?

5.3 Apply Jensen's inequality to $f(x) = x^2$. What familiar fact do you get?

5.4 Find the mode of S_T (the peak of its density) by differentiating the lognormal density, and evaluate it.

5.5 The put payoff $\max(K - s, 0)$ is convex. Use Jensen's inequality to give a lower bound for $E[\max(K - S_T, 0)]$. Then compute the exact value, using $\max(K - s, 0) = \max(s - K, 0) - s + K$ and Lesson 4's formula for the call.

5.6 A year's log return is the sum of 252 independent daily log returns, each with mean $0.01/252$ and standard deviation $0.2/\sqrt{252}$, but with an unknown distribution. Use the central limit theorem to approximate the probability that the year's log return exceeds 0.2.

5.7 *(Code)* Simulate S_T with and without the $-\frac{1}{2}\sigma^2$, for $\sigma = 20\%$ and $\sigma = 40\%$. Compare each average with $S_0 e^{rT}$.

5.8 *(Code)* Replace the coin-flip days with days whose log returns are uniform with the same mean and variance. Compare the 1% and 99% quantiles of S_T with the lognormal ones.

Self-test

1. Why is the median of S_T below its mean?
2. If the volatility doubles, what happens to $E[S_T]$?
3. Which is bigger, $E[\ln S_T]$ or $\ln E[S_T]$, and why?

Companion notebook: `s5e05_lognormal_jensen.ipynb`. Every number in this chapter is computed by it.

LESSON 6 · VIDEO 6

Convergence and Inequalities

What it means for a random quantity to converge; Markov's and Chebyshev's inequalities, which hold for every distribution; the mean and variance of an average and the $\sqrt{n}$ law; the law of large numbers, which makes Monte Carlo work; and the argument "mean to a constant, variance to zero", which shows that squared random steps add up to time.

Why this lesson

Two promises from earlier lessons still need proofs.

1. **Monte Carlo works.** Many lessons check a formula by averaging simulated payoffs, assuming the average of many simulations settles on the true expectation. Why should it, and how many simulations are enough?
2. **Squared random steps add up to time.** Lesson 1 claimed that random steps of size $\sigma\sqrt{\Delta t}$ have squares that add up to something that does not vanish. Lesson 14 needs that to be a theorem.

Both rest on the same small toolkit: two inequalities (Markov and Chebyshev), the law of large numbers, and one convergence argument. This lesson builds them, proving every step.

What you need from earlier lessons

- **Expectation and variance** (Lesson 3). Linearity; $\mathrm{Var}(aX) = a^2\mathrm{Var}(X)$; independent variances add; the indicator $\mathbf{1}\{A\}$.
- **The normal** (Lesson 4). $E[Z^2] = 1$, $E[Z^4] = 3$, $E|Z| = \sqrt{2/\pi}$, and $N(1.96) = 0.975$.
- **The lognormal share** (Lesson 5). $E[S_T] = S_0 e^{rT}$ and its standard deviation; the central limit theorem.

The picture

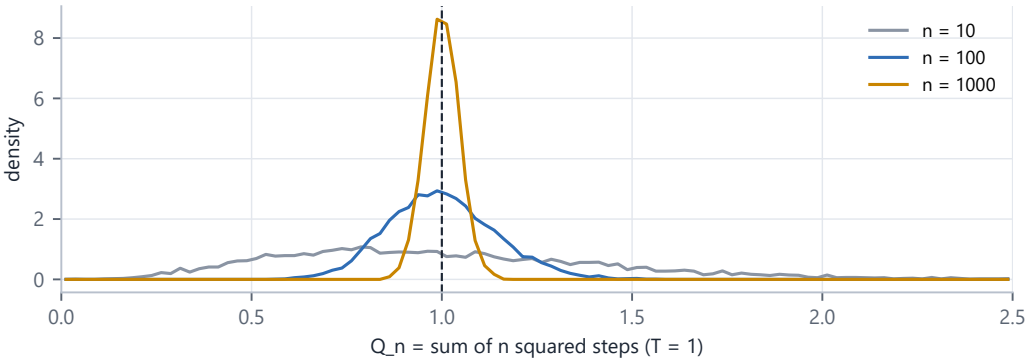


Figure 6.1. The sum of n squared random steps, each normal with variance 1/n. Its average is always 1; as n grows, its spread shrinks around 1. Mean fixed, variance to zero: the sum converges to 1.

Every curve is centred on 1, and the curves get narrower as n grows. In the end the random sum is pinned at 1. That is the whole idea of this lesson: **if the mean stays put and the variance goes to zero, the random quantity converges to the mean.**

What does "converge" mean for random quantities?

For ordinary numbers, $x_n \to c$ means x_n eventually stays as close to c as you like. A random quantity X_n cannot be forced that close on *every* outcome: there is always some small chance of an unlucky draw. So we ask instead that **the chance of being far away goes to zero**.

Definition. X_n **converges in probability** to c if, for every $\varepsilon > 0$ (however small),

$$P\big(|X_n - c| \geq \varepsilon\big) \longrightarrow 0 \quad \text{as } n \to \infty.$$

Read it as: pick any tolerance ε; the probability of missing c by more than ε shrinks to nothing.

Two related ideas appear in this course:

- **Convergence in mean square:** $E[(X_n - c)^2] \to 0$. It is stronger: it implies convergence in probability (shown below). Lesson 14 uses it for Brownian motion.
- **Convergence in distribution** (Lesson 5's central limit theorem): probabilities of intervals converge, but the random quantity itself need not settle anywhere.

To prove convergence in probability we need a way to bound probabilities like $P(|X_n - c| \geq \varepsilon)$. That is what the next two inequalities do.

Markov's inequality

Theorem (Markov). If $Y \geq 0$ and $a > 0$, then

$$P(Y \geq a) \leq \frac{E[Y]}{a}.$$

Intuition. If the average income in a town is 50,000, at most 10% of people can earn 500,000 or more: if more did, they alone would push the average above 50,000.

Proof, in two steps.

Step 1. An inequality that holds outcome by outcome. Compare Y with $a\,\mathbf{1}\{Y \geq a\}$, which is a when $Y \geq a$ and 0 otherwise.

- If $Y \geq a$: the right side is a, and $Y \geq a$.
- If $Y < a$: the right side is 0, and $Y \geq 0$.

Either way, $Y \geq a\,\mathbf{1}\{Y \geq a\}$.

Step 2. Take expectations. Averaging preserves the inequality, and the expectation of an indicator is the probability of its event:

$$E[Y] \geq a\,E\big[\mathbf{1}\{Y \geq a\}\big] = a\,P(Y \geq a).$$

Divide by a. □

Example. The share price is never negative and has mean $E[S_T] = 103.05$. So the chance it ends at twice its mean or more, $S_T \geq 206.09$, is at most 0.50. The actual log-normal probability is 1.8e-04. Markov's inequality is very cautious, because it uses only the mean. It is valuable because it **always** holds.

Chebyshev's inequality

Theorem (Chebyshev). If X has mean μ and variance v, then for every $\varepsilon > 0$,

$$P\big(|X - \mu| \geq \varepsilon\big) \leq \frac{v}{\varepsilon^2}.$$

Equivalently, with $\varepsilon = k\,\mathrm{sd}(X)$: the chance of landing k or more standard deviations from the mean is at most $1/k^2$.

Proof.

Step 1. Square. Both sides of $|X - \mu| \geq \varepsilon$ are non-negative, so it holds exactly when $(X - \mu)^2 \geq \varepsilon^2$:

$$P\big(|X - \mu| \geq \varepsilon\big) = P\big((X - \mu)^2 \geq \varepsilon^2\big).$$

Step 2. Markov. Apply Markov's inequality to $Y = (X - \mu)^2$, which is ≥ 0, with $a = \varepsilon^2$:

$$P\big((X-\mu)^2 \geq \varepsilon^2\big) \leq \frac{E[(X-\mu)^2]}{\varepsilon^2} = \frac{v}{\varepsilon^2}. \square$$

How good is it? Chebyshev uses only the variance, so it holds for every distribution with a finite variance, and is loose for any particular one. For the lognormal share of Lesson 5:

Chance of landing at least k standard deviations from the mean. Lognormal column: two million simulated prices.

K STANDARD DEVIATIONS	CHEBYSHEV'S BOUND	ACTUAL, LOGNORMAL SHARE	IF IT WERE NORMAL
1.5	0.4444	0.1214	0.1336
2.0	0.2500	0.0426	0.0455
3.0	0.1111	0.0068	0.0027
4.0	0.0625	0.0011	0.0001

At two standard deviations, Chebyshev promises at most 25%; the truth is about 4%. But the promise needs no knowledge of the distribution, and for proving that something converges, that is exactly what we want.

The average of many draws

Let $X_1, X_2, \ldots, X_n$ be independent, each with mean μ and variance v (for example, n simulated share prices). Their **average** is

$$\bar{X}_n = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

Its mean. By linearity,

$$E[\bar{X}_n] = \frac{1}{n}\big(E[X_1] + \cdots + E[X_n]\big) = \frac{n\mu}{n} = \mu.$$

Its variance. Pull out the constant $1/n$ (it comes out squared), then use independence (variances add):

$$\operatorname{Var}(\bar{X}_n) = \frac{1}{n^2}\operatorname{Var}(X_1 + \cdots + X_n) = \frac{1}{n^2} \cdot nv = \frac{v}{n}.$$

The $\sqrt{n}$ law. So the standard deviation of the average is

$$\operatorname{sd}(\bar{X}_n) = \frac{\operatorname{sd}(X)}{\sqrt{n}}.$$

This is the most important fact about simulation: **the error shrinks like $1/\sqrt{n}$.** To halve it you need four times as many draws; to gain one more decimal digit, a hundred times as many.

The law of large numbers

Theorem (weak law of large numbers). With $X_1, X_2, \dots$ as above, $\bar{X}_n \to \mu$ in probability: for every $\varepsilon > 0$,

$$P\big(|\bar{X}_n - \mu| \ge \varepsilon\big) \longrightarrow 0.$$

Proof. Apply Chebyshev to $\bar{X}_n$, which has mean μ and variance v/n:

$$P\big(|\bar{X}_n - \mu| \ge \varepsilon\big) \le \frac{v/n}{\varepsilon^2} = \frac{v}{n\,\varepsilon^2} \longrightarrow 0. \ \square$$

That is why Monte Carlo works: an average of many independent simulated payoffs converges to the true expectation.

Monte Carlo in action. Averages of simulated year-end prices, against the exact $E[S_T] = 103.0455$ (the standard deviation of one price is 20.8169):

The error is usually within one or two multiples of sd/√n, and that multiple shrinks like 1/√n.

PATHS	AVERAGE	ERROR	SD/√N
100	104.1261	1.0807	2.0817
1,000	103.3468	0.3014	0.6583
10,000	102.8604	-0.1850	0.2082
100,000	103.0538	0.0083	0.0658
1,000,000	103.0440	-0.0015	0.0208

To check the $\sqrt{n}$ law directly, the notebook computes 200 independent averages of 1,000 prices each. They scatter with a standard deviation of 0.6491, against the predicted $20.82/\sqrt{1000} = 0.6583$.

How many paths are enough?

Suppose we want the Monte Carlo average within 0.10 of $E[S_T]$ with probability at least 95%.

With Chebyshev (a guarantee). We need $P(|\bar{X}_n - \mu| \ge 0.1) \le 0.05$. Chebyshev bounds the left side by $v/(n \times 0.1^2)$, so it is enough that

$$\frac{v}{n \times 0.01} \le 0.05 \quad \Longleftrightarrow \quad n \ge \frac{v}{0.05 \times 0.01} = 866,687,$$

using $v = 433.34$.

With the central limit theorem (an estimate). Lesson 5 says $\bar{X}_n$ is close to normal with standard deviation $\mathrm{sd}/\sqrt{n}$. A normal is within 1.96 standard deviations of its mean 95% of the time, so we need $1.96\,\mathrm{sd}/\sqrt{n} = 0.1$:

$$n = \left(\frac{1.96\,\mathrm{sd}}{0.1}\right)^2 = 166,467.$$

Chebyshev asks for 5.2 times as many paths. It is a guarantee for *every* distribution, so it is cautious for this one. In practice we use the normal approximation and report $\bar{X}_n \pm 1.96\,\mathrm{sd}/\sqrt{n}$ as a 95% interval.

Mean to a constant, variance to zero

The law of large numbers is one example of a general argument that Part 3 uses again and again.

Proposition. If $E[X_n] \to c$ and $\mathrm{Var}(X_n) \to 0$, then $X_n \to c$ in probability.

Proof. Fix $\varepsilon > 0$.

Step 1. The mean is eventually close. Since $E[X_n] \to c$, for all large n we have $|E[X_n] - c| < \varepsilon/2$.

Step 2. Being far from c forces being far from the mean. The distance from X_n to c is at most the distance from X_n to $E[X_n]$ plus the distance from $E[X_n]$ to c (the **triangle inequality**: a detour is never shorter):

$$|X_n - c| \le |X_n - E[X_n]| + |E[X_n] - c| < |X_n - E[X_n]| + \tfrac{\varepsilon}{2}.$$

So if $|X_n - c| \ge \varepsilon$, then $|X_n - E[X_n]| > \varepsilon/2$.

Step 3. Chebyshev. Therefore

$$P\big(|X_n - c| \ge \varepsilon\big) \le P\big(|X_n - E[X_n]| \ge \tfrac{\varepsilon}{2}\big) \le \frac{\mathrm{Var}(X_n)}{(\varepsilon/2)^2} = \frac{4\,\mathrm{Var}(X_n)}{\varepsilon^2} \longrightarrow 0. \qquad \square$$

The mean-square version. Write $X_n - c = (X_n - E[X_n]) + (E[X_n] - c)$, square, and take expectations. The cross term has expectation 0, because $E[X_n - E[X_n]] = 0$. So

$$E\big[(X_n - c)^2\big] = \mathrm{Var}(X_n) + \big(E[X_n] - c\big)^2 \longrightarrow 0.$$

The same two conditions give convergence in mean square too. (And mean square implies in probability: by Markov, $P(|X_n - c| \ge \varepsilon) = P\big((X_n - c)^2 \ge \varepsilon^2\big) \le E[(X_n - c)^2]/\varepsilon^2$.)

The example that matters: squared steps add up to time

Let $Y_1, \dots, Y_n$ be independent $N(0, T/n)$: n random steps, each with variance T/n. Part 3 will call them the steps of a Brownian path over a time T. Their **sum of squares** is

$$Q_n = Y_1^2 + Y_2^2 + \cdots + Y_n^2.$$

Write each step as a scaled standard normal, $Y_i = \sqrt{T/n}\, Z_i$, so $Y_i^2 = \frac{T}{n} Z_i^2$.

Step 1. The mean. By linearity, with $E[Z_i^2] = 1$:

$$E[Q_n] = \sum_{i=1}^{n} \frac{T}{n} E[Z_i^2] = n \cdot \frac{T}{n} = T.$$

Step 2. The variance of one term. Using Lesson 4's $E[Z^4] = 3$:

$$\mathrm{Var}(Z^2) = E[Z^4] - \big(E[Z^2]\big)^2 = 3 - 1 = 2, \qquad \mathrm{Var}\Big(\frac{T}{n} Z_i^2\Big) = \frac{T^2}{n^2} \times 2.$$

Step 3. The variance of the sum. The terms are independent, so their variances add:

$$\mathrm{Var}(Q_n) = n \times \frac{2T^2}{n^2} = \frac{2T^2}{n} \longrightarrow 0.$$

Step 4. Conclude. The mean stays at T and the variance goes to zero, so by the proposition $Q_n \to T$ in probability (and in mean square): **the squared steps add up to the elapsed time**, and the randomness disappears in the limit. In simulation, with $T = 1$:

5,000 simulated sets of n steps for each n.

STEPS	AVERAGE OF Q	ITS SD	√(2/N)	SUM OF |STEPS|	√(2N/Π)
10	1.0034	0.4433	0.4472	2.53	2.52
100	1.0007	0.1424	0.1414	7.98	7.98
1,000	1.0010	0.0445	0.0447	25.24	25.23
10,000	1.0003	0.0143	0.0141	79.80	79.79

This is the fact Lesson 1 promised: random steps of size $\sqrt{\Delta t}$ have squares that add up to time, not to zero. Lesson 14 calls Q_n's limit the **quadratic variation** of Brownian motion, and Itô's lemma (Lesson 16) rests on it.

Squared steps settle, absolute steps blow up

The last two columns of the table show the opposite behaviour for the **absolute** steps. By Lesson 4, $E|Z| = \sqrt{2/\pi}$, so $E|Y_i| = \sqrt{T/n}\,\sqrt{2/\pi}$ and

$$E\Big[\sum_{i=1}^{n}|Y_i|\Big] = n\sqrt{\frac{T}{n}}\sqrt{\frac{2}{\pi}} = \sqrt{\frac{2nT}{\pi}} \longrightarrow \infty.$$

The total length of the path, the sum of its absolute steps, grows without limit, while the sum of squares settles at T.

Why the difference. Each absolute step is of size $\sqrt{T/n}$, and n of them give about $n\sqrt{T/n} = \sqrt{nT}$, which grows. Each squared step is of size T/n, and n of them give T. For a smooth path it is the other way round: steps of size $1/n$ add up to a finite length, and their squares add up to about $n/n^2 \to 0$ (Lesson 1). A random path is so rough that its length is infinite but its quadratic variation is finite. Lesson 14 builds on exactly this contrast.

What to remember

- $X_n \to c$ in probability: $P(|X_n - c| \geq \varepsilon) \to 0$ for every $\varepsilon > 0$.
- Markov: $Y \geq 0 \Rightarrow P(Y \geq a) \leq E[Y]/a$. Chebyshev: $P(|X - \mu| \geq \varepsilon) \leq \mathrm{Var}(X)/\varepsilon^2$.
- The average of n independent draws has mean μ and standard deviation $\mathrm{sd}/\sqrt{n}$.
- Law of large numbers: $\bar{X}_n \to \mu$. Monte Carlo error shrinks like $1/\sqrt{n}$; report $\bar{X}_n \pm 1.96\,\mathrm{sd}/\sqrt{n}$.
- Mean $\to c$ and variance $\to 0 \Rightarrow X_n \to c$ (in probability and in mean square).
- For independent $N(0, T/n)$ steps: $\sum Y_i^2 \to T$, while $E\sum|Y_i| = \sqrt{2nT/\pi} \to \infty$.

Common mistakes

- **"Chebyshev needs a normal distribution."** It needs only a finite variance; that is why it is loose.
- **"Twice the paths halves the error."** The error falls like $1/\sqrt{n}$: halving it takes four times the paths.
- **Forgetting that $1/n$ comes out of a variance squared.** $\mathrm{Var}(\bar{X}_n) = \frac{1}{n^2}\mathrm{Var}(\sum X_i)$, not $\frac{1}{n}\mathrm{Var}(\sum X_i)$.
- **"Converges in probability" means "equals the limit for large n".** It means the chance of being more than ε away shrinks to zero; any single outcome can still be unlucky.
- **"Second moment" and "quadratic variation" are the same.** $E[Y_i^2]$ is a number for each step; $Q_n = \sum Y_i^2$ is a random sum along one path. The second moments tell you Q_n's mean; the convergence argument tells you Q_n itself stops being random.

Exercises

Solutions are in *Part 1: Solutions*.

6.1 Prove Markov's inequality, $P(Y \geq a) \leq E[Y]/a$ for $Y \geq 0$ and $a > 0$, and deduce Chebyshev's inequality from it.

6.2 Let $Y_1, \dots, Y_n$ be independent $N(0, T/n)$ and $Q_n = \sum Y_i^2$. Show $E[Q_n] = T$ and $\mathrm{Var}(Q_n) = 2T^2/n$.

6.3 With the same Y_i, show $E\big[\sum |Y_i|\big] = \sqrt{2nT/\pi}$.

6.4 How many Monte Carlo paths does Chebyshev say you need for $|\bar{X}_n - E[S_T]| < 0.10$ with probability at least 95%? How many does the normal approximation suggest?

6.5 The share price $S_T \geq 0$ has mean $S_0 e^{rT} = 103.05$. What does Markov's inequality say about $P(S_T \geq 150)$? Use the lognormal distribution of Lesson 5 to compute the actual probability.

6.6 Let $X_n = n$ with probability $1/n$ and $X_n = 0$ otherwise. Show that $X_n \to 0$ in probability, although $E[X_n] = 1$ for every n. What happens to $\mathrm{Var}(X_n)$?

6.7 *(Code)* Estimate $P(S_T > K)$ by simulation for $n = 10^3, 10^4, 10^5, 10^6$, with a 95% interval each time, and compare with $N(d_-)$ from Lesson 5.

6.8 *(Code)* Simulate Q_n and $\sum |Y_i|$ for n up to 10^5 and confirm the formulas of Exercises 6.2 and 6.3.

Self-test

1. Does Chebyshev's inequality need the normal distribution?
2. If $E[X_n] = 2 + 1/n$ and $\mathrm{Var}(X_n) = 1/n$, what does X_n converge to?
3. Why does $\sum |Y_i|$ blow up while $\sum Y_i^2$ settles?

Companion notebook: `s5e06_convergence.ipynb`. Every number in this chapter is computed by it.

Formula sheet and glossary

Formula sheet

Calculus (Lessons 1 and 2)

- Chain rule: $\big(f(g(x))\big)' = f'(g(x))\,g'(x)$; along a path, $\frac{d}{dt}g(t,x(t)) = g_t + g_x\,x'(t)$.
- Taylor: $f(a+h) = f(a) + f'(a)h + \frac{1}{2}f''(a)h^2 + O(h^3)$.
- Two variables: $\Delta g \approx g_t\Delta t + g_x\Delta x + \frac{1}{2}\big(g_{tt}\Delta t^2 + 2g_{tx}\Delta t\,\Delta x + g_{xx}\Delta x^2\big)$.
- Substitution: $\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du$; by parts: $\int_a^b uv'\,dx = [uv]_a^b - \int_a^b u'v\,dx$.
- Series: $e^x = \sum_k x^k/k!$; $\ln(1+h) = h - h^2/2 + h^3/3 - \dots$
- Euler's method: $m \leftarrow m + km\,\Delta t$, error proportional to Δt.
- $m' = km \Rightarrow m(t) = m(0)e^{kt}$; $(1+r/n)^n \to e^r$.

Probability (Lesson 3)

- $E[g(X)] = \int g(x)f(x)\,dx$; $E[aX+bY+c] = aE[X] + bE[Y] + c$.
- $\mathrm{Var}(X) = E[X^2] - E[X]^2$; $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$.
- $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$; $\rho = \mathrm{Cov}(X,Y)/(\mathrm{sd}\,X\,\mathrm{sd}\,Y)$.
- Independent: $E[XY] = E[X]E[Y]$, and variances add.

The normal distribution (Lesson 4)

- $X = \mu + sZ$: $P(X \le x) = N\big((x-\mu)/s\big)$; $N(-x) = 1 - N(x)$.
- $M_X(u) = E[e^{uX}]$, $M_X^{(k)}(0) = E[X^k]$; independent sums: $M_{X+Y} = M_X M_Y$.
- $E[e^{uZ}] = e^{u^2/2}$; for any normal X, $E[e^X] = e^{\text{mean}+\frac{1}{2}\text{variance}}$.
- $E[Z^2] = 1$, $E[Z^4] = 3$, $E[Z^6] = 15$; $E|Z| = \sqrt{2/\pi}$.
- Independent normals: means add, variances add.
- Completing the square: $e^{uz}\varphi(z) = e^{u^2/2}\varphi(z-u)$, so $E[e^{uZ}\mathbf{1}\{Z>a\}] = e^{u^2/2}N(u-a)$.

Lognormal prices (Lesson 5), with $S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma\sqrt{T}Z}$:

- $E[S_T] = S_0 e^{rT}$; $E[S_T^2] = S_0^2 e^{(2r+\sigma^2)T}$; $\mathrm{Var}(S_T) = S_0^2 e^{2rT}(e^{\sigma^2 T} - 1)$.
- Density: $f(s) = \dfrac{1}{s\,\sigma\sqrt{T}}\,\varphi\Big(\dfrac{\ln(s/S_0) - (r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\Big)$ for $s > 0$.

- Median $S_0 e^{(r-\frac{1}{2}\sigma^2)T}$; mode $S_0 e^{(r-\frac{3}{2}\sigma^2)T}$.
- $P(S_T > K) = N(d_-)$; $e^{-rT}E[S_T \mathbf{1}\{S_T > K\}] = S_0 N(d_+)$; $d_\pm = \dfrac{\ln(S_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$.
- Jensen: $E[f(X)] \geq f(E[X])$ for convex f.

Convergence (Lesson 6)

- Markov: $P(Y \geq a) \leq E[Y]/a$; Chebyshev: $P(|X - \mu| \geq \varepsilon) \leq \mathrm{Var}(X)/\varepsilon^2$.
- $\mathrm{Var}(\bar{X}_n) = \mathrm{Var}(X)/n$; $E[X_n] \to c$ and $\mathrm{Var}(X_n) \to 0 \Rightarrow X_n \to c$.
- Y_i independent $N(0, T/n)$: $\sum Y_i^2 \to T$; $E\sum |Y_i| = \sqrt{2nT/\pi} \to \infty$.

The course's numbers

Share price today, S_0	100
Strike, K	105
Risk-free rate, r (continuous)	3%
Volatility, σ	20%
Horizon, T	1 year

Glossary

Central limit theorem. A standardised sum of many independent, identically distributed pieces with finite variance is close to normal.

Chain rule. The derivative of a composition is the product of the derivatives, each at the right point.

Chebyshev's inequality. $P(|X - E[X]| \geq k\,\mathrm{sd}) \leq 1/k^2$, for any distribution with finite variance.

Completing the square. Rewriting $uz - \frac{1}{2}z^2$ as $-\frac{1}{2}(z-u)^2 + \frac{1}{2}u^2$; it turns $e^{uz}\varphi(z)$ into a shifted normal density.

Continuous compounding. Growth by e^{rT}: the limit of compounding n times a year as $n \to \infty$.

Convergence in probability. $X_n \to c$ if $P(|X_n - c| \geq \varepsilon) \to 0$ for every $\varepsilon > 0$.

Convex function. One that bends upwards ($f'' \geq 0$): it lies above its tangent lines. Concave is the opposite.

Correlation. Covariance divided by both standard deviations; a number between −1 and 1.

Covariance. $E[XY] - E[X]E[Y]$; zero for independent variables, but zero covariance does not imply independence.

Cross term. The $2g_{tx}\Delta t\,\Delta x$ term in the two-variable Taylor expansion.

Density. A function $f \geq 0$ with total area 1 whose areas are probabilities.

Derivative. The slope of a curve at a point, $f'(x) = \lim_{h\to 0}\big(f(x+h) - f(x)\big)/h$; the rate of change.

Distribution function. $F(x) = P(X \leq x)$; its derivative is the density.

Expectation. The probability-weighted average of a random variable.

Independence. Knowing one random variable tells you nothing about the other: $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$.

Integrating factor. Multiplying $m' = km$ (or $m' = km + c$) by e^{-kt} to make the left side an exact derivative.

Integration by parts. The product rule read backwards: $\int uv' = [uv] - \int u'v$.

Jensen's inequality. $E[f(X)] \geq f(E[X])$ for convex f.

Law of large numbers. The average of many independent copies of a random variable settles on its expectation.

Lognormal distribution. The distribution of e^Y for normal Y.

Markov's inequality. $P(Y \geq a) \leq E[Y]/a$ for $Y \geq 0$.

Median, mode. The value with half the probability below it; the peak of the density.

Moment. $E[X^k]$, the k-th moment: mean ($k = 1$), spread ($k = 2$), lopsidedness ($k = 3$), tail weight ($k = 4$).

Moment generating function. $M(u) = E[e^{uX}]$; its power-series coefficients are $E[X^k]/k!$, so its derivatives at 0 are the moments.

Normal distribution. $N(\mu, s^2)$, with density $\frac{1}{s\sqrt{2\pi}}e^{-(x-\mu)^2/(2s^2)}$.

Partial derivative. A derivative in one variable with the others held fixed. For an option: delta V_S, gamma V_{SS}, theta V_t.

Quadratic variation. The limit of the sum of squared steps of a path; for random steps of variance T/n it is T.

Random variable. A number attached to each outcome of a random experiment.

Standard deviation. The square root of the variance, in the same units as the random variable.

Substitution. The chain rule read backwards; change the limits along with the variable.

Taylor's theorem. A smooth function equals its Taylor polynomial plus a remainder of the next order.

Variance. $E[(X - E[X])^2] = E[X^2] - E[X]^2$.

Further reading

- Joseph K. Blitzstein and Jessica Hwang, *Introduction to Probability* (free online): Lessons 3 to 6, in depth.
- Thomas Mikosch, *Elementary Stochastic Calculus with Finance in View*, Chapter 1: a compact probability primer aimed at this course's destination.
- Steven E. Shreve, *Stochastic Calculus for Finance I*, Chapter 1 and the appendices: where Part 2 of this course begins.